

Galaxy Image Classification and the Galaxy Merger Rate

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We train a convolutional neural network (CNN) to predict Galaxy Zoo 2 (GZ2) morphological vote fractions from Sloan Digital Sky Survey (SDSS) *gri* images. Starting from 61,578 labelled galaxies with 37 vote-fraction labels in $[0, 1]$, we centre-crop and resize the 424×424 cutouts to 96×96 pixels, a $\sim 20\times$ reduction in pixel count, and split 80/20 into 49,262 training and 12,316 validation images. A mean-prediction baseline achieves a bulk validation RMSE of 0.1639 and a mean per-label validation RMSE of 0.1406. We compare two backbones, a 4-block custom CNN with dropout and ~ 2.8 M parameters and an 18-layer ResNet adapted for small inputs and sigmoid output with ~ 11.2 M parameters, and apply two optimisations (**ReduceLROnPlateau** learning-rate scheduling and random $[0, 360^\circ)$ rotation augmentation exploiting the rotational invariance of galaxy morphology) together with one loss-level regularisation (tree-weighted RMSE following S. Dieleman et al. 2015). We report performance under both bulk RMSE (cell-equal weighting) and mean per-label RMSE (label-equal weighting); the latter is our headline metric because it neutralises the cell-equal-weighting reward for predicting near-zero on the many rare deep-tree cells whose ground truth concentrates near zero. Under that metric the augmented Custom CNN trained with the tree-weighted RMSE loss attains the best mean per-label validation RMSE of 0.0883, a 37% improvement over the mean-prediction baseline. Per-label scatter plots show that top-level smooth/disk features are predicted accurately, while residuals concentrate in the rare odd-feature leaves (rings, mergers, lens/arcs). Applied to 79,975 held-out SDSS images, the tree-weighted model recovers a mean-probability merger fraction $f_{\text{merger}} = 7.23\% \pm 0.04\%$ on Class8.6, converting to a major-merger rate $R_{\text{merger}} \approx 0.07\text{--}0.36 \text{ Gyr}^{-1}$ over the $\tau_{\text{vis}} \approx 0.2\text{--}1.1$ Gyr observability range of J. M. Lotz et al. (2008, 2011) and V. Rodriguez-Gomez et al. (2015). The mean-probability estimator overshoots the $z \approx 0$ Lotz value of $\sim 0.02 \text{ Gyr}^{-1}$ by a factor of $\sim 3\text{--}18$ because the tree-weighted loss provides no gradient signal on Class8.6 for ordinary (non-odd) galaxies, leaving the network free to predict diffusely-positive values rather than near zero; restricting to high-confidence predictions ($p > 0.4\text{--}0.5$) gives $f_{\text{merger}} = 1\text{--}3\%$, bracketing the Lotz literature value within the τ_{vis} systematic.

1. Introduction

Galaxy morphology is the most direct observational tracer of a galaxy’s dynamics and history (C. J. Conselice 2014). Over large samples, morphology allows us to reconstruct the statistical evolution of the galaxy population without following individual galaxies through cosmic time. In the hierarchical Λ CDM model, the primary mechanism evolving galaxies along its evolutionary sequence are major mergers, where they perturb stellar orbits, drive gas to galactic centres fuelling starbursts and AGN, and convert rotationally-supported discs into pressure-supported spheroids. The major-merger rate as a function of redshift therefore sets the rate at which the early-Universe disc population evolves into the present-day red sequence. Constraining this rate observationally is required for a quantitative theory of galaxy evolution. Locally, this rate has been constrained at the $\sim 0.02\text{--}0.03 \text{ Gyr}^{-1}$ level by observation (J. M. Lotz et al. 2011) and simulation (V. Rodriguez-Gomez et al. 2015), but the two methodologies disagree on the $(1+z)$ evolution slope by roughly a factor of two.

The largest hand-labelled morphological catalogue available is Galaxy Zoo 2 (GZ2; K. W. Willett et al.

2013), comprising $\sim 3 \times 10^5$ SDSS galaxies labelled by citizen-scientist consensus on a 37-leaf decision tree. However, as this labelling pipeline struggle to scale to the LSST, Euclid, and Roman data releases with its reliance on voluntary participation, convolutional neural networks (CNNs) provide a programmatic alternative where given paired images and labels, a CNN learns to predict morphological vote fractions directly from pixels (A. Krizhevsky et al. 2017).

In this work, we (1) assemble a 61,578-galaxy GZ2 training set with 96×96 pixel SDSS *gri* cutouts and 37 vote-fraction labels per galaxy; (2) train two CNN architectures, an 18-layer ResNet (~ 11.2 M parameters; K. He et al. 2015) and a custom 4-block CNN (~ 2.8 M parameters); (3) compare two optimisations (learning-rate scheduling and 360° rotation augmentation) and one loss-level regularisation (tree-weighted RMSE following S. Dieleman et al. 2015) against a mean-prediction baseline; (4) apply the best model to a held-out test sample of 79,975 SDSS galaxies; and (5) recover a major-merger rate R_{major} from the predicted Class8.6 merger fraction using the J. M. Lotz et al. (2011) visibility-timescale calibration.

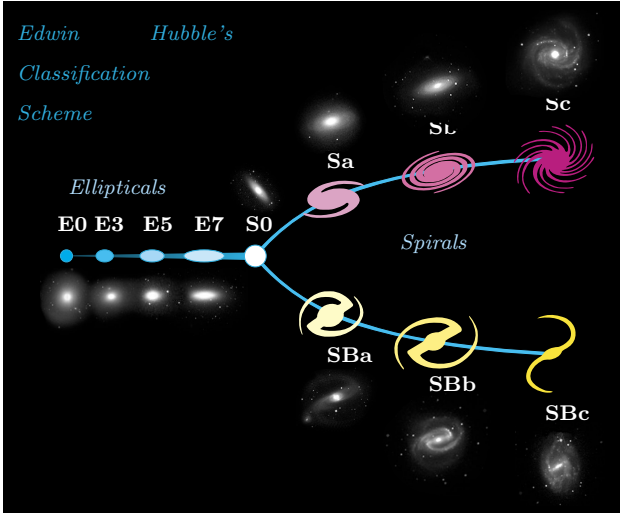


Fig. 1.— The Hubble tuning-fork diagram. Ellipticals (E0–E7) lie along the handle, ordered by apparent flattening. The two prongs separate barred (SBa–SBc) from unbarred (Sa–Sc) spirals along an axis of decreasing bulge prominence and increasingly open arms. Lenticulars (S0) bridge the elliptical and disk families. Image credit: NASA & ESA, <https://esahubble.org/images/heic9902o/>.

2. Background

2.1. The Hubble Sequence

The Hubble sequence (Figure 1) separates galaxies into two physically distinct branches: pressure-supported ellipticals (E0–E7) and lenticulars (S0), and rotationally-supported spirals (Sa–Sc) and irregulars (Irr). The Sa–Sc sequence within the disk class is ordered by decreasing bulge-to-total ratio (B/T), increasing gas fraction, and increasing star-formation rate per unit mass. Ellipticals are thought to form through major mergers that scramble ordered rotation into a velocity-dispersion-supported spheroid, while disk galaxies assemble more quiescently through cold gas accretion and angular-momentum conservation (C. J. Conselice 2014).

2.2. The Cosmic Merger Rate

The major-merger rate per galaxy had conventionally been measured by two independent methods. J. M. Lotz et al. (2011) count morphologically disturbed systems using non-parametric indicators including the Gini coefficient, M_{20} (the second-order moment of light at the 20th-percentile brightness), CAS (Concentration, Asymmetry, Smoothness), and multi-mode/intensity/deviation $\{M, I, D\}$ statistics. They convert disturbed fractions into rates using a simulation-calibrated visibility timescale τ_{vis} , the time during which a major merger remains morphologically disturbed¹. Applied to close-pair counts in mass-

selected SDSS samples, the morphological method gives a power-law redshift evolution

$$R_{\text{major}}(z) = R_0 (1+z)^\alpha, \quad (1)$$

with $R_0 = 0.03 \pm 0.01 \text{ Gyr}^{-1}$ and $\alpha = +1.7 \pm 1.3$ for the J. M. Lotz et al. (2011) close-pair fit. V. Rodriguez-Gomez et al. (2015) measure the same rate by counting subhalo mergers in the Illustris cosmological hydrodynamical simulation, yielding $\alpha = +2.6 \pm 0.2$ for the Milky-Way-mass. At $z \approx 0.07$ this gives $R_{\text{major}} \approx 0.03\text{--}0.04 \text{ Gyr}^{-1}$, the local-universe reference value we compare against in Section 6.4.

The two $z \approx 0$ normalisations agree to within $\sim 1.5\times$, but the redshift-evolution exponents differ by roughly a factor of two. Both methods are indirect, with the morphological method converting a disturbed-galaxy fraction into a rate through τ_{vis} , and the simulation method reading merger events off a subhalo tree of which realised mapping onto observable galaxies is itself not well understood.

2.3. The Galaxy Zoo 2 Catalogue

A complementary, more direct approach would be to instead ask classifiers, human or machine, whether a galaxy visibly looks disturbed or merging, and aggregate the resulting fractions across a flux- and resolution-limited sample.

2.3.1. Sample selection and imaging

GZ2 (K. W. Willett et al. 2013) solicits classifications from $\sim 84,000$ volunteer classifiers over ~ 14 months, totalling ~ 16 million votes on 304,122 SDSS DR7 Legacy galaxies selected with Petrosian magnitude $m_r < 17.0$, Petrosian half-light radius $r_{\text{Petro},90} > 3''$, and redshift $0.0005 < z < 0.25$ from the SDSS spectroscopic Main Galaxy Sample. The selection imposes the same flux and apparent-size cuts as GZ1 but is restricted to the brightest, best-resolved $\sim 34\%$ of the GZ1 sample so that deeper morphological questions (bars, arm count, dust lanes) can be more unambiguously identified.

Each galaxy is presented as an SDSS *gri* colour-composite JPEG cutout, scaled per-galaxy at $0.024 r_{\text{Petro},90}$ arcseconds per pixel and rendered at a fixed 424×424 pixels. The pixel scale therefore tracks each galaxy’s physical extent rather than its on-sky size with angular footprints varying from cutout to cutout (roughly $10''$ for the smallest objects up to $\sim 2'$ for the largest), while the physical footprint is approximately $\sim 1.7 r_{\text{Petro},90}$ on a side, corresponding to $\sim 12 \text{ kpc}$ at the median GZ2 redshift $z \approx 0.07$.

2.3.2. Classification scheme

Classifiers answer an 11-question decision tree (Figure 2): Q1 (smooth/features/star), Q2 (edge-on?), Q3 (bar?), Q4 (spiral?), Q5 (bulge prominence), Q6 (anything odd?), Q7 (how round?), Q8 (odd feature:

orbital parameters, mass ratio, and gas fraction.

¹J. M. Lotz et al. (2008) calibrate τ_{vis} against hydrodynamic merger simulations and find $\tau_{\text{vis}} \approx 0.2\text{--}1.1 \text{ Gyr}$ depending on

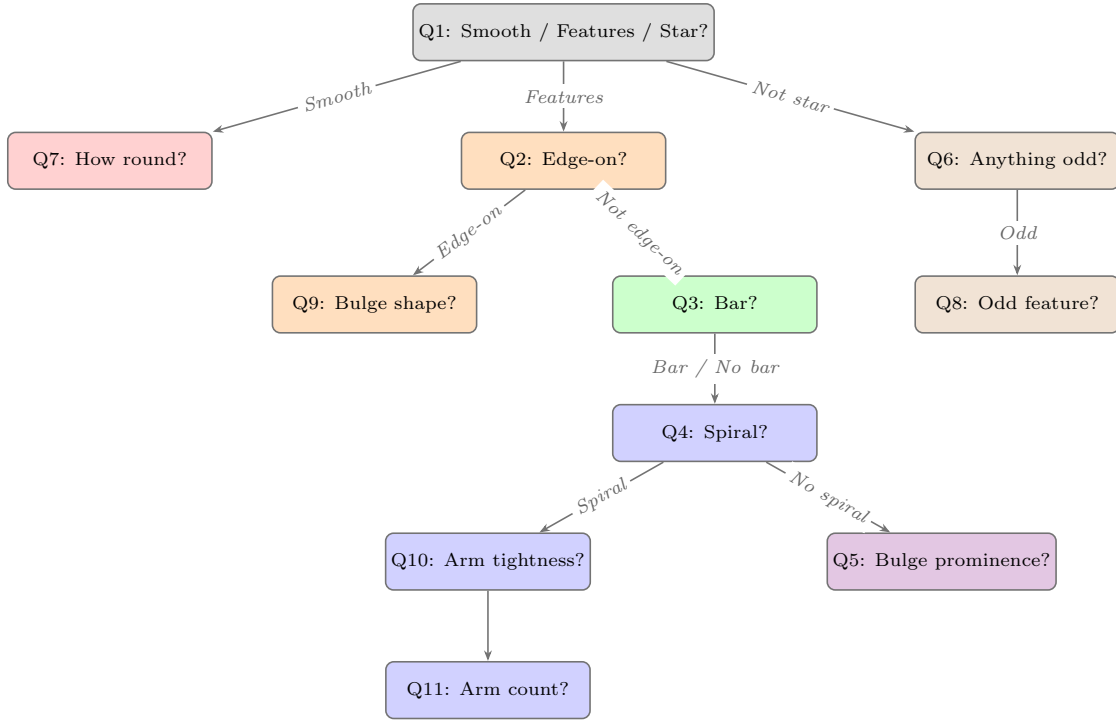


Fig. 2.— GZ2 hierarchical decision tree. Each node is one of the 11 GZ2 questions; directed edges show which downstream questions are gated by each answer at the parent node (e.g. Q7 “How round?” is only asked when a classifier first chose “Smooth” at Q1). The 37 leaf answers, the vote fractions we regress against, sit on the leaves of this tree (not shown for brevity). Node colours denote branch: gray = root, red = smooth, orange = edge-on, green = face-on/bar, blue = spiral, violet = bulge, brown = odd-feature.

ring, lens/arc, disturbed, irregular, dust lane, merger, other), Q9 (edge-on bulge shape), Q10 (arm tightness), and Q11 (arm count). Each question is gated by a previous answer. For example, Q7 is asked only when Q1 = smooth, Q9 only when Q2 = yes, and so on. The 37 leaf answers form a hierarchical tree rather than a flat vector, and the reported labels are the fraction of classifiers who selected each leaf.

This tree expands substantially on the flat six-category GZ1 scheme of C. J. Lintott et al. (2008) (elliptical, clockwise spiral, anticlockwise spiral, edge-on disk, star/don’t-know, merger). GZ2 drops the spiral chirality split, which proved susceptible to a documented left–right perception bias in GZ1 (K. Land et al. 2008), and adds eight new questions (Q2, Q3, Q5, Q7, Q8–Q11) for a sixfold increase in morphological resolution (K. W. Willett et al. 2013).

2.3.3. Vote aggregation and cleaning

A typical galaxy receives 40–50 independent classifications, with a minimum of ~ 15 . Galaxies are shown to classifiers in randomised order, drawn from a queue that prioritises objects with the fewest existing classifications. To handle classifier noise and resolution-driven bias, K. W. Willett et al. (2013) apply three cleaning steps: (i) classifier-consistency weighting, in which each user’s vote weight is iteratively downweighted by their disagreement with the running consensus on a calibration sub-sample, suppressing the contribution of careless or adversarial

classifiers; (ii) debiasing of vote fractions against apparent magnitude M_r and angular radius R_{50} following the method of S. P. Bamford et al. (2009); and (iii) parent-normalisation of child vote fractions to enforce the parent-sum consistency of the decision tree.

Together, the catalogue reports, for each question q , the classifier-consistency-weighted and resolution-debiased fraction $f_q \in [0, 1]$ of classifiers who selected that answer (we measure $\sim 55,000$ distinct values across our 61,578 training labels for Class1.1). Sibling fractions within a question sum to the parent’s vote fraction by construction. By reporting fractions rather than hard majority labels, we preserve classifier-disagreement since an ambiguous galaxy at $f_q \approx 0.5$ would be distinguishable from a confident one at $f_q \approx 1.0$.

2.3.4. Classification bias and apparent redshift evolution

For a galaxy with some morphology, at higher redshifts, it subtends a smaller angle, has lower surface brightness for resolved features, and is broadened by the SDSS $\sim 1.4''$ point-spread function. Spiral arms, bars, and bulge–disk features that are easily identified in a nearby galaxy are less resolved at $z \gtrsim 0.1$, so classifiers preferentially answer “smooth” and “round” for distant objects, surfacing as an apparent redshift evolution in mean GZ2 morphological type (S. P. Bamford et al. 2009).

K. W. Willett et al. (2013) correct for this following S. P. Bamford et al. (2009) by binning the sample in narrow redshift slices (where the intrinsic morphology mix is assumed fixed) and measuring how f_{feature} , f_{bar} , etc. depend on M_r and R_{50} within each slice; every galaxy’s raw vote fraction is then projected onto a well-resolved reference baseline ($M_r \approx -20.17$, $R_{50} \approx 3''$) at which the bias is by definition zero. Uniquely, the merger fraction f_{merger} cannot be recovered through this since at higher redshift, neighbouring galaxies blend into the target’s PSF and tidal features become harder to resolve, biasing the merger vote fraction upward rather than downward, so high- z galaxies are preferentially classified as mergers via the same resolution effect that pushes other features toward “smooth”. Therefore, while we use the corrected vote fractions as our labels, a residual resolution-dependent bias remains in the merger-probabilities because the intrinsic rate also scales as $(1+z)^\alpha$ (J. M. Lotz et al. 2011; V. Rodriguez-Gomez et al. 2015).

2.3.5. Comparison to other morphological catalogues

Compared to alternative morphological catalogues, GZ2 has a larger sample size, containing $\sim 304,122$ galaxies, $\sim 20\times$ the P. B. Nair & R. G. Abraham (2010) (NA10) expert catalogue. Per-galaxy uncertainty is quantified by the ~ 40 independent votes rather than a single non-parametric scalar such as the de Vaucouleurs T-type (running from -6 for ellipticals to $+10$ for irregulars) or a Gini- M_{20} value. K. W. Willett et al. (2013) show that GZ2 vote fractions correlate strongly with NA10 T-types, so the citizen-science labels carry the same morphological information as expert classifications at a scale expert pipelines cannot match.

However, the drawback of any visual identification approach, be it expert T-typing, citizen-science voting, is that it returns a subjective answer where a non-parametric statistic such as G , M_{20} , CAS, or $\{M, I, D\}$ returns a deterministic scalar. The same pixels run through the Lotz pipeline produce the same Gini coefficient every time; the same pixels presented to two GZ2 classifiers, or to the same classifier on two different days, can produce different votes. Three galaxies from our own classification session on the GZ2 interface (<https://www.zooniverse.org/projects/zookeeper/galaxy-zoo/>) demonstrates this subjectivity (Figure 3).

The middle and bottom panels are more unambiguous, depicting a centrally concentrated smooth galaxy and a resolved spiral (2 spiral arms), on which essentially any classifier, human or non-parametric, could mostly return the same label. In contrast, the top panel depicts a bright bulge-like core embedded in low-surface-brightness extended emission for which the “spiral / not spiral” and “edge-on disk with dust lane” identifications could be visually degenerate at SDSS resolution, and the resulting vote fraction would be split. A non-parametric estimator would instead

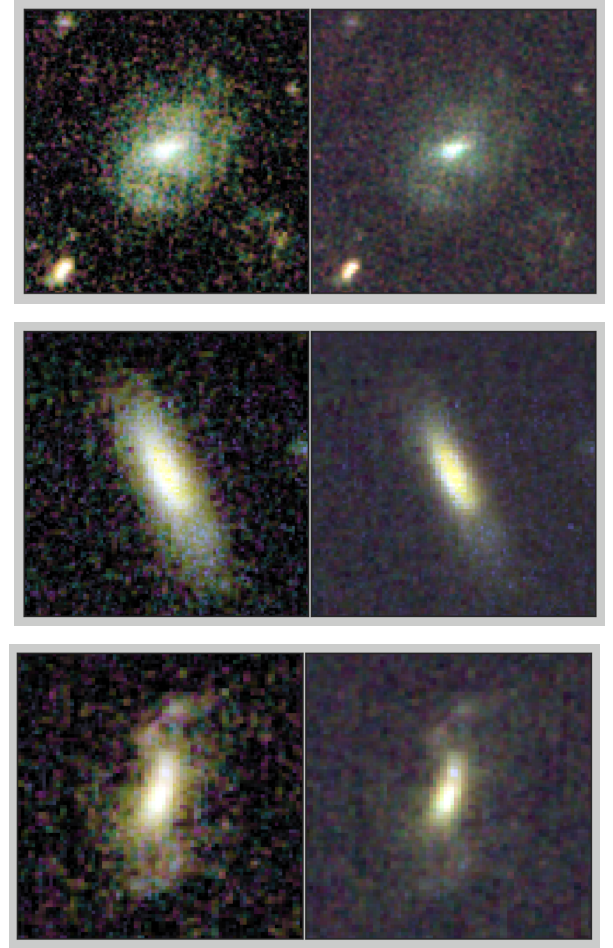


Fig. 3.— Three GZ2 galaxies from our own classification session. Top: an ambiguous case with a bright bulge-like core embedded in low-surface-brightness extended emission, where the Q4 “spiral / not spiral” and Q8 “dust lane” axes are visually degenerate at SDSS resolution, so the GZ2 vote fraction sits at intermediate values. Middle: a clean “smooth” galaxy with centrally concentrated mass, red, no resolved features. Bottom: a less-clean but visually discernible spiral galaxy with a clearly resolved disk with visible arm structure.

collapse such a galaxy to a deterministic scalar, while instead discarding the multi-feature information (e.g., bar presence, arm count, bulge prominence, oddity type) that GZ2 resolves across 37 separate leaves. A CNN trained on GZ2 would thus inherit both sides, where it operates at GZ2’s full 37-axis resolution, while propagating the per-image subjective noise that human classifiers contributed to the labels.

3. Data

The training set consists of 61,578 SDSS *gri* composite images (424×424 pixels) and 37 GZ2 classification vote fraction labels in $[0, 1]$ per galaxy. Figure 4 shows a random montage of 28 galaxies drawn from the training set.

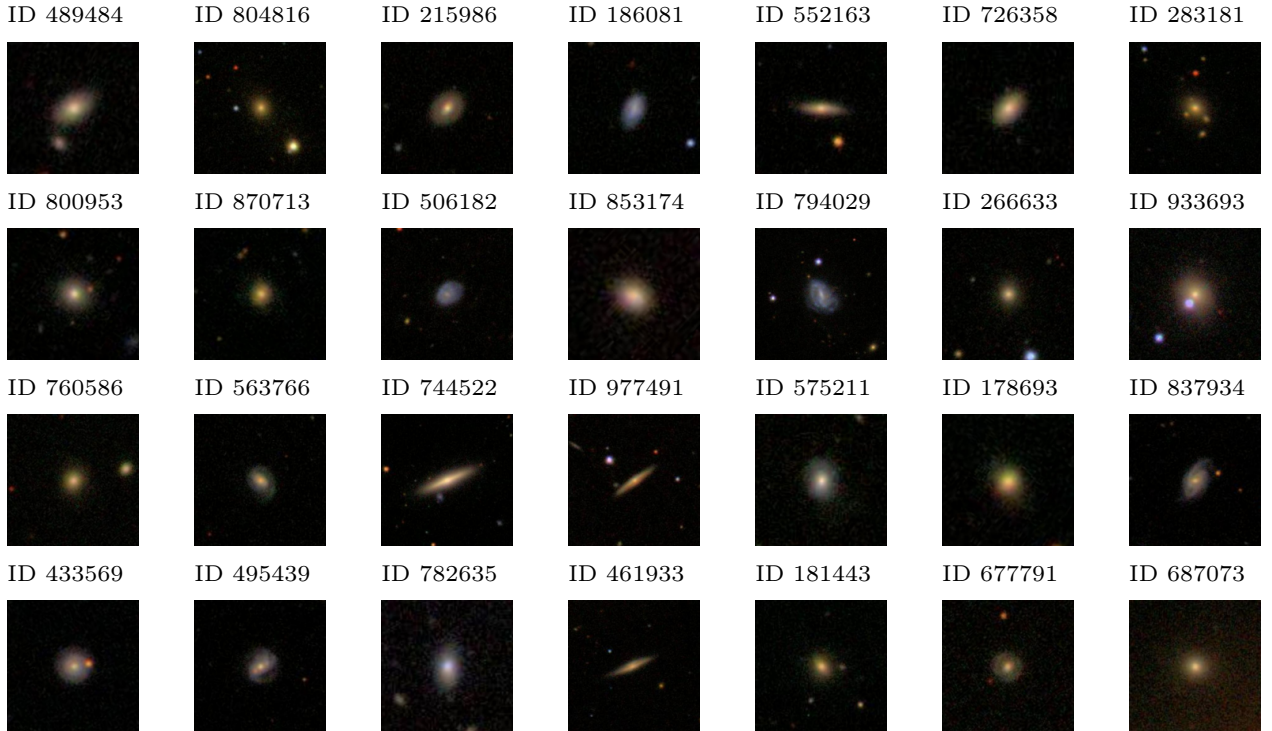


Fig. 4.— 28 random training-set galaxies displayed at full 424×424 resolution. Galaxy IDs are annotated in each panel. The sample spans the full GZ2 morphological range from smooth red ellipticals to blue grand-design spirals, with $\sim 20\%$ of objects showing edge-on disks, bars, or interaction features. The colour bimodality between red-sequence ellipticals and blue-cloud disks is visually obvious at the pixel level, supporting the morphological colour information discussed in Section 3.1.

3.1. SDSS *gri* Composite Images

We use SDSS DR7 (K. N. Abazajian et al. 2009) three-colour cutouts in the *gri* bands, rendered as RGB JPEGs via the R. Lupton et al. (2004) scaling: *g* ($\lambda_{\text{eff}} \approx 477$ nm) to blue, *r* ($\lambda_{\text{eff}} \approx 623$ nm) to green, and *i* ($\lambda_{\text{eff}} \approx 763$ nm) to red. This mapping reproduces the colour-magnitude bimodality between blue-cloud star-forming disks ($g - r \sim 0.4$) and red-sequence quiescent ellipticals ($g - r \sim 0.8$ at $z \sim 0$). As such, both human visual and machine CNN approaches can leverage pixel colour as a proxy for top-level Smooth/Features distinction; the harder labels (bar, arm count, dust lane, odd features) require morphological structure labelling beyond the dominant colour gradient.

The GZ2 sample spans $z \sim 0.02$ – 0.25 with median $z \approx 0.07$, drawn from the SDSS spectroscopic main galaxy sample (M. A. Strauss et al. 2002; C. Stoughton et al. 2002). The native SDSS pixel scale is $0.396''/\text{pixel}$, corresponding to ~ 0.54 kpc/pixel at the median redshift assuming $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The GZ2 cutouts are pre-scaled per galaxy so that the field of view tracks $\sim 1.7 r_{\text{Petro},90}$ on each side such that each image reproduce the physical scale and extent of each galaxy.

3.1.1. Prototype Images

Figure 5 shows the prototype image (highest vote fraction) for each label. Visually obvious labels (smooth, edge-on disk, grand-design 2-arm spiral) have clean, morphologically distinctive prototypes. Subtle or rare labels (lens/arc Class8.2, dust lane Class8.7, high-arm-count spirals Class11.3–11.5) require detecting faint, narrow, or rare features that are easily confused with PSF artefacts or anomalous disk structure.

3.2. Label Distributions

Figure 6 shows normalized histograms of all 37 labels. The top-level labels (Class1.1/1.2 and Class6.1/6.2) are broad and roughly bimodal between its members, with distribution heaviest at 0 or 1, reflecting strong classifier consensus on the smooth-vs-disk distinction. The conditional sub-class labels (Class7.x, Class10.x, and Class11.x) are instead concentrated near zero because most galaxies were never asked the corresponding gated question. The rare-feature labels (Class1.3 star/artefact and the Class8.x odd-feature leaves) are also sharply peaked at zero but with a short positive tail, reflecting the smaller distribution of astrophysical anomalies: rings, lens/arcs, dust lanes, and mergers each occur in only a few percent of the sample.

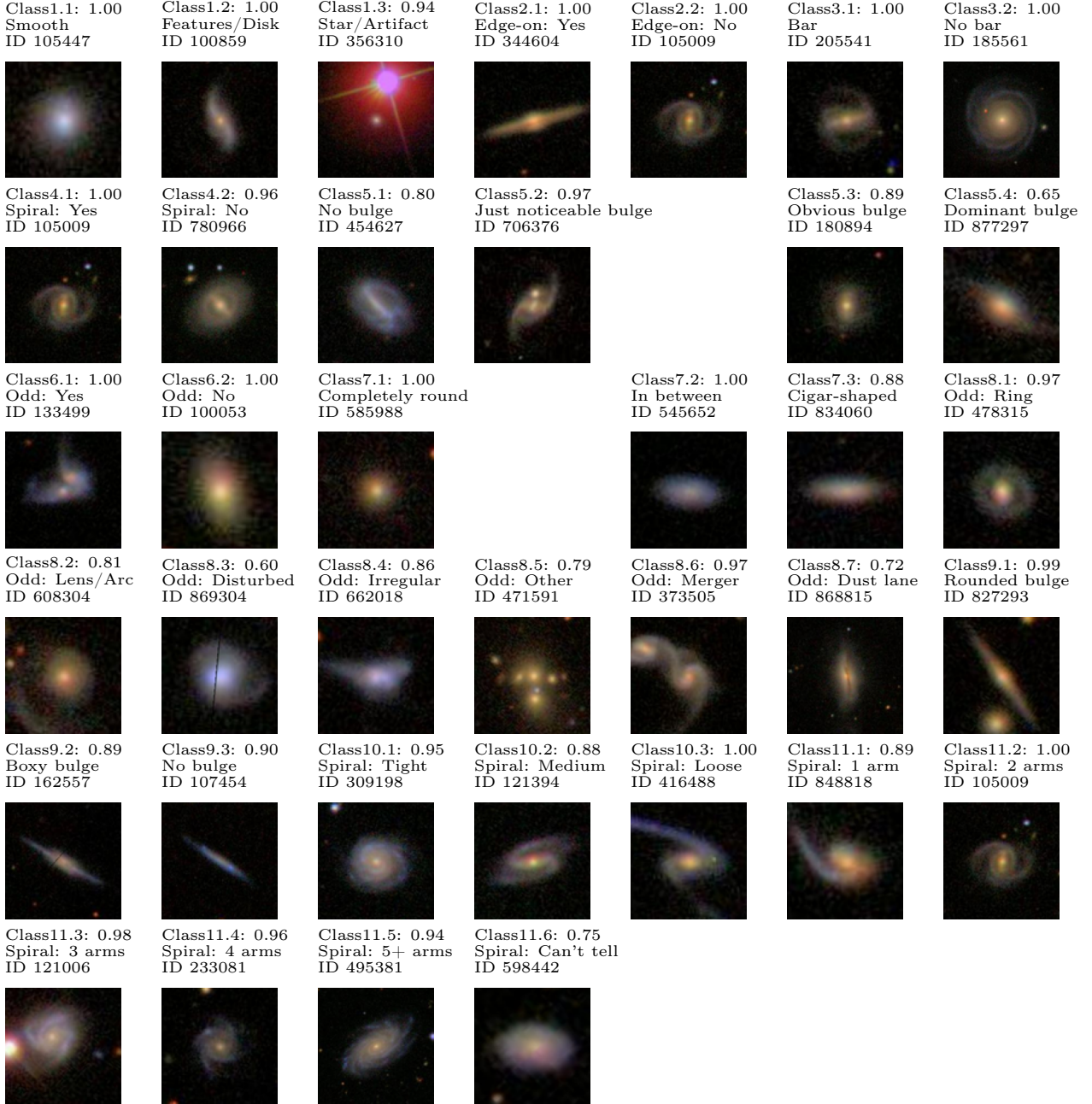


Fig. 5.— Prototype images: for each of the 37 GZ2 labels, the galaxy in the training set that maximises the corresponding vote fraction, displayed at the model-input crop and labelled by the descriptive class name. Visually obvious labels (smooth, edge-on disk, grand-design $m=2$ spiral, no-bulge) have clean and morphologically distinctive prototypes, whereas the deep-tree rarer labels (lens/arc, dust lane, high arm count) exhibit non-unique/varying or low-contrast prototypes foreshadow a higher per-label-RMSE pattern of Section 5.

3.2.1. Label Correlations

We compute the 37×37 Pearson correlation matrix (Figure 7, reordered by tree branch to make the block structure visible) using

$$\rho_{ij} = \frac{\langle ij \rangle - \langle i \rangle \langle j \rangle}{\sqrt{\langle i^2 \rangle - \langle i \rangle^2} \sqrt{\langle j^2 \rangle - \langle j \rangle^2}}. \quad (2)$$

For any k -way question with perfect classifier consensus, we would expect sibling answers to have correlations $\rho = -1/(k-1)$. For the effectively-binary

Q1 (Class1.1 vs Class1.2; Class1.3 is near-zero everywhere, so Q1 collapses to a two-way vote in practice) and Q6 (Class6.1 vs Class6.2) we measure $\rho = -0.992$ and $\rho = -1.000$ respectively, matching the binary expectation of a strong anticorrelation -1 . For the $k = 4$ bulge-prominence question Q5 we measure -0.20 (for roughly similar labels e.g., no-bulge and just barely noticeable) to -0.40 (no-bulge and dominant), near the expected $-1/3$. Parents and childs are generally strongly correlated.

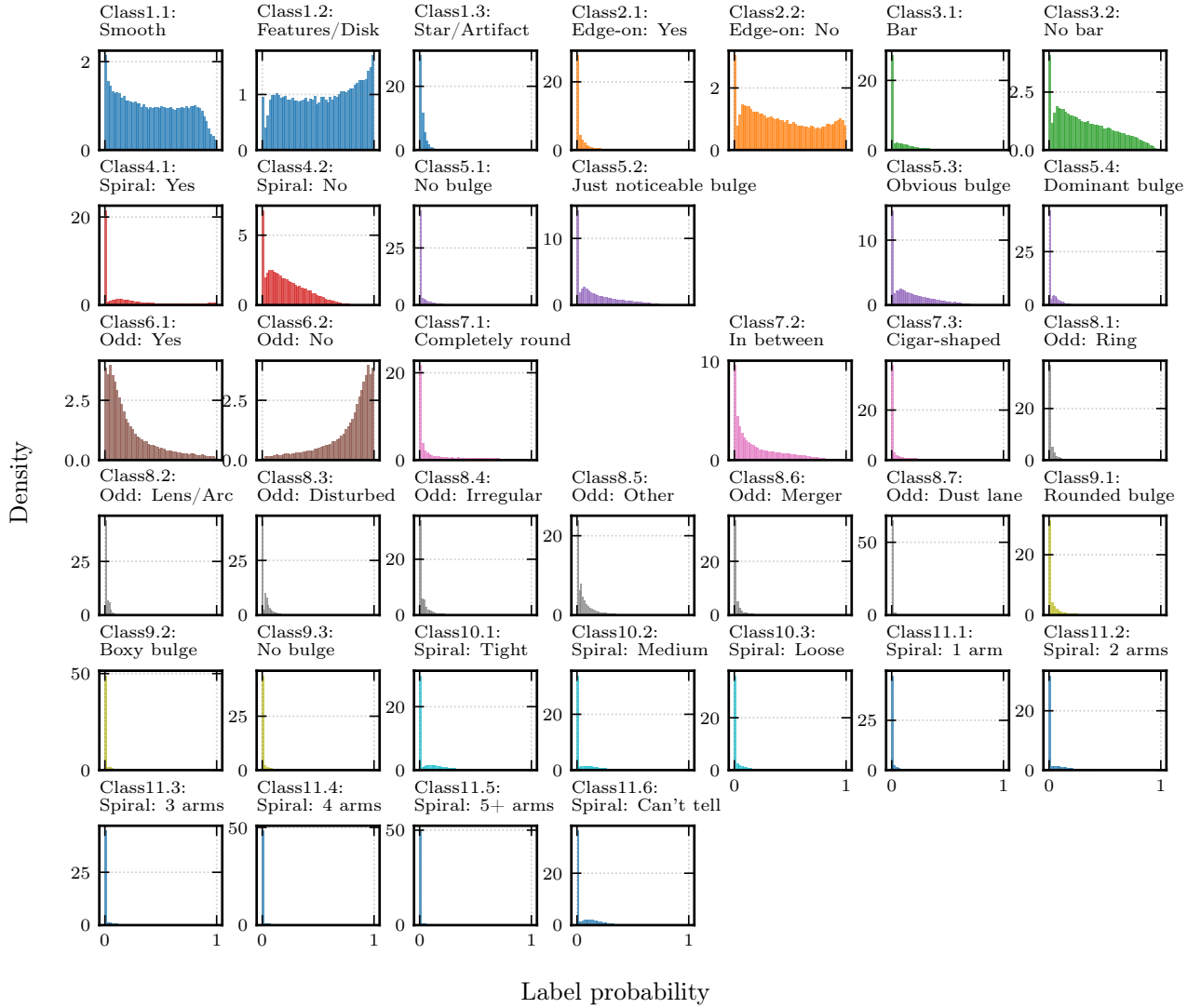


Fig. 6.— Normalized histograms of all 37 GZ2 classification labels. Top-level labels show broad distributions; deeper hierarchical labels are concentrated near zero because most galaxies were never asked the corresponding gated question. Histograms are colored in enumeration by class.

The seven odd-feature labels (Class8.1–Class8.7: ring, lens/arc, disturbed, irregular, other, merger, dust lane) correlate only weakly with each other, with measured pairwise $|\rho| \lesssim 0.1$ in most cases. Apart from some suppression stemming from $\rho_{\text{sibling}} = -1/(k-1) = -1/6 \approx -0.17$ for perfect classifier consensus (i.e., they are mutually exclusive), the joint distribution of any two odd labels is overwhelmingly dominated by negative pairs due to their rarities (and thus significant peaks at 0). As such, the measured anti-correlations are systematically weaker than the theoretical $-1/6$.

For the asymmetries, features/disk markedly correlates more strongly with edge-on no ($\rho = +0.748$) than with edge-on yes ($\rho = +0.333$), directly tracing the $\cos i$ inclination distribution of randomly-oriented thin disks where edge-on views ($i > 80^\circ$) subtend only $\sim 17\%$ of the solid angle, so most disk galaxies are face-on or moderately inclined.

The Cigar-shaped label (Class7.3) is anomalous. While nominally a smooth sub-label, it correlates positively with Edge-on Yes ($\rho = +0.496$) and with the edge-on bulge labels Class9.1 ($\rho = +0.370$) and Class9.3 ($\rho = +0.475$), it is uncorrelated with Smooth itself ($\rho = -0.001$). Highly inclined disks are therefore systematically misclassified as “smooth + cigar-shaped” when the disk substructure is unresolved at SDSS seeing, a confusion the CNN inherits from the labels.

3.3. Image Preprocessing

Raw GZ2 cutouts are $424 \times 424 \times 3$ in *gri* RGB, most of which is empty sky. Per image this is ~ 0.54 MB in `uint8` and ~ 2.16 MB in `float32`, totalling ~ 33 GB for the 61,578-galaxy training set in `uint8` and ~ 133 GB in `float32`. Both are well above the 12 GB VRAM of a local GPU and the 24 GB system RAM available. The pipeline therefore:

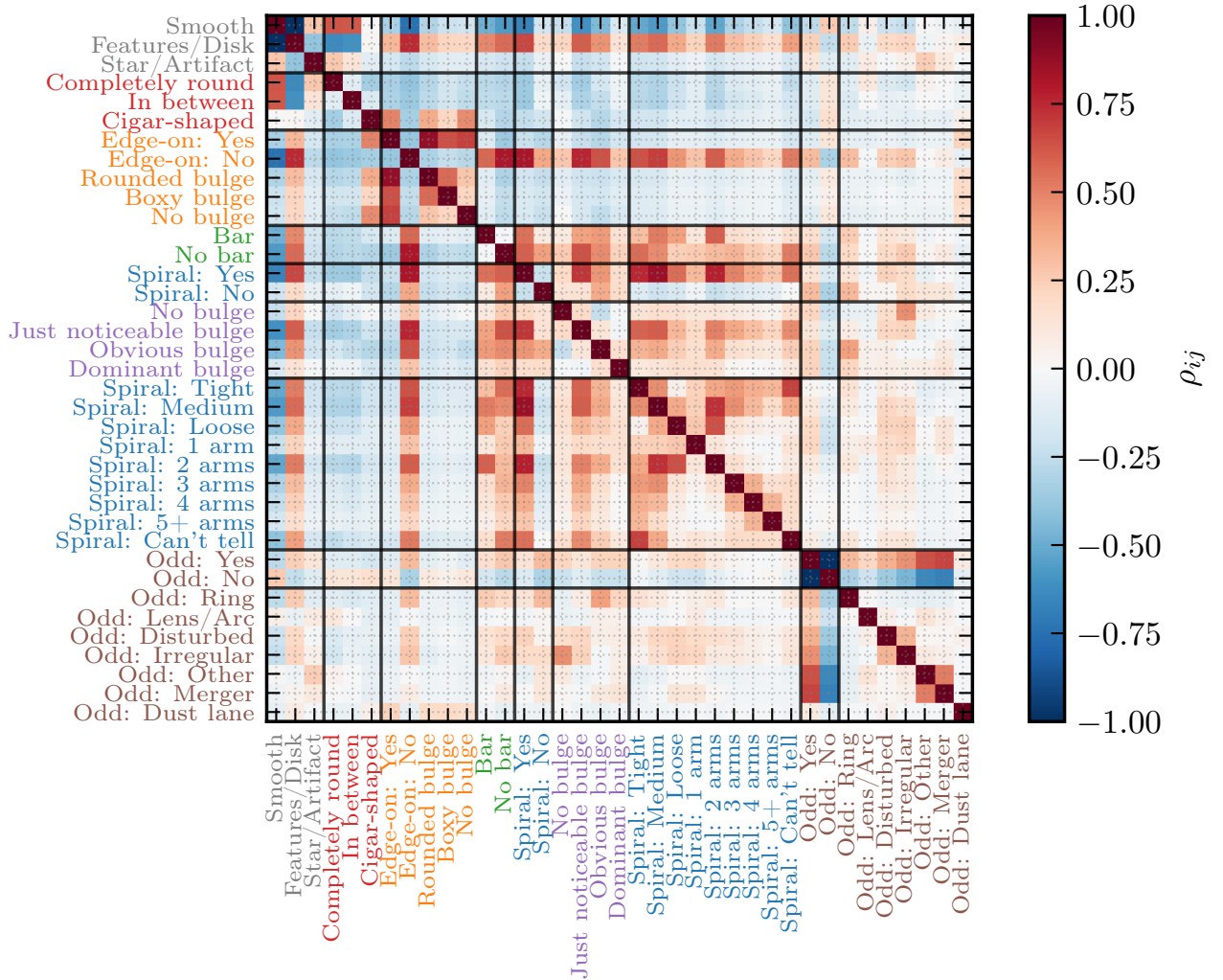


Fig. 7.— 37×37 label correlation matrix reordered by tree branch (smooth / edge-on / face-on bar-spiral / odd-feature). Block-diagonal structure aligns with the hierarchy, making sibling question negative correlations and parent–child positive correlations visually separable; off-diagonal cross-branch correlations are dominated by being in different subtrees, otherwise they are weakly or uncorrelated.

1. Centre-crops by 15% per side (retaining the central $\sim 296^2$ region around the galaxy).
2. Bilinearly resizes to a 136×136 on-disk cache in uint8 (~ 3.4 GB total).
3. GPU-side `CenterCrop(96)` at training time yields the $96 \times 96 \times 3$ model input.

A 2-step downscaling process is opted for to avoid black corners due to rotation and avoid additional memory requirements of full resolution images. However, this comes with the trade-off of potential compression artifacts and loss of detail. The full pipeline (raw $424^2 \rightarrow 136^2$ cache $\rightarrow$ rotated 96^2 model input) is graphically depicted in the top row of Figure 10; Figure 8 shows the same four-stage pipeline applied to four example galaxies, verifying that the rotation-safe cache buffer eliminates corner artefacts at every rotation angle.

We chose a slightly larger image resolution 96×96

input to preserve image quality rather than memory. Table 1 lists the worst-case pixel scales for a small galaxy at the lower end of the GZ2 selection ($r_{\text{Petro},90} \approx 3''$; K. W. Willett et al. 2013). Because each GZ2 cutout is pre-scaled per galaxy at $0.024 \times r_{\text{Petro},90}$ arcsec/pixel, every galaxy fills a similar fraction of its 424^2 frame regardless of on-sky angular size.

At this resolution floor, prominent morphological feature scales of interest barely fit within the 96^2 field of view. Spiral arm widths (~ 0.5 – 2 kpc; C. J. Conselice 2014) span ~ 10 – 30 pixels, bar lengths (~ 2 – 8 kpc; J. A. L. Aguerri et al. 2009) span ~ 30 – 100 pixels, bulge effective radii ($r_e \sim 0.5$ – 2 kpc; D. A. Gadotti 2009) span ~ 10 – 30 pixels, and edge-on dust lanes ($h_z \sim 100$ – 300 pc; J. J. Dalcanton et al. 2004) span ~ 2 – 5 pixels at the resolvability limit at $\sim 1.4''$ FWHM.

Stage	Angular	Physical (at $z = 0.07$)
Native SDSS imaging	$0.396''/\text{pix}$	$\sim 540 \text{ pc}/\text{pix}$
GZ2 per-galaxy cutout (424^2)	$\sim 0.020''/\text{pix}$	$\sim 27 \text{ pc}/\text{pix}$
Model input (96^2)	$\sim 0.040''/\text{pix}$	$\sim 60 \text{ pc}/\text{pix}$
SDSS PSF FWHM	$\sim 1.4''$	$\sim 1.9 \text{ kpc}$

Table 1: Nested pixel scales of the pipeline at the median GZ2 redshift $z \approx 0.07$, evaluated for a small galaxy at the lower end of the GZ2 selection ($r_{\text{Petro},90} \approx 3''$) where the per-galaxy pixel scale is finest. The model input field of view spans $\sim 6 \text{ kpc}$ on a side and the SDSS point-spread-function FWHM is the fundamental resolution limit. Larger galaxies in the GZ2 sample have proportionally coarser pixel scales (up to a factor of $\sim 3\text{--}5$), so the values quoted here serve as an extrema check of the resolution floor.

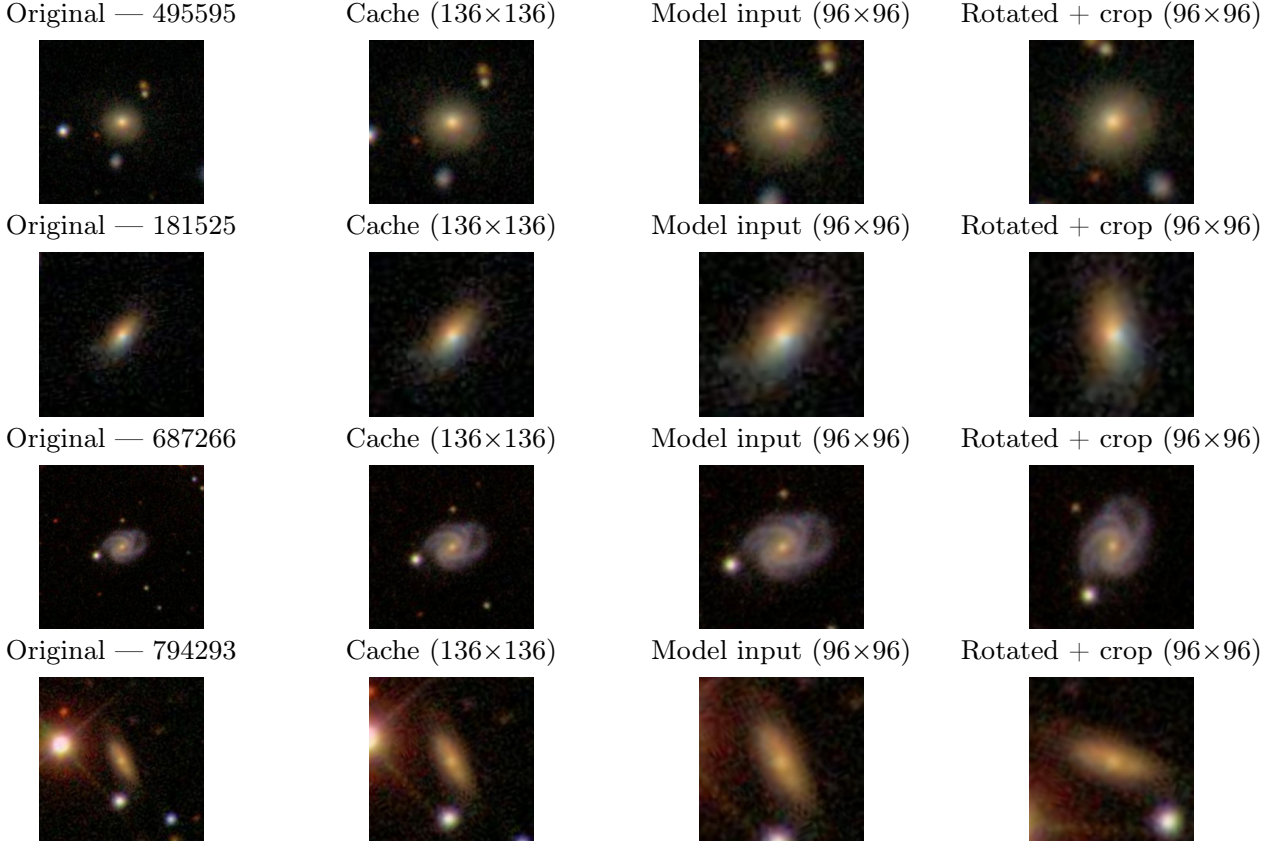


Fig. 8.— Pipeline stages for four example galaxies (rows): raw 424^2 SDSS GZ2 cutout, 136^2 rotation-safe cache, 96^2 centre-cropped model input, and one realisation of the same image after a random rotation by $\theta \in [0, 360^\circ)$. The rotated image always covers the centre 96^2 region with no black corner artefacts because the cache size $136 \geq 96\sqrt{2}$ is the smallest integer guaranteeing rotation-safety.

3.4. Training/Validation Split

We randomly split the dataset 80/20 into 49,262 training and 12,316 validation images. Figure 9 confirms that the per-label distributions are statistically indistinguishable between the two sets.

3.5. Baseline Model

The underlying principle we adopt in training the models is to minimise the root-mean-square error (RMSE) between predicted and true vote-fraction ma-

trices,

$$L_{\text{RMSE}} = \sqrt{\frac{1}{N_{\text{gal}} N_{\text{lab}}} \sum_{i=1}^{N_{\text{gal}}} \sum_{j=1}^{N_{\text{lab}}} (\ell_{ij}^{\text{true}} - \ell_{ij}^{\text{pred}})^2}, \quad (3)$$

where $\ell_{ij}^{\text{true}}, \ell_{ij}^{\text{pred}} \in [0, 1]$ are the true and predicted vote fractions for galaxy i and label j , and $N_{\text{lab}} = 37$.

We adopt a baseline model which predicts the training-set mean label vector for every image, i.e. $\ell_{ij}^{\text{pred}} = \bar{\ell}_j^{\text{train}}$ independent of the input image, where $\bar{\ell}_j^{\text{train}}$ is the per-label average over the training set. This yields training and validation RMSEs of 0.1638 and 0.1639 respectively. This serves as the RMSE floor

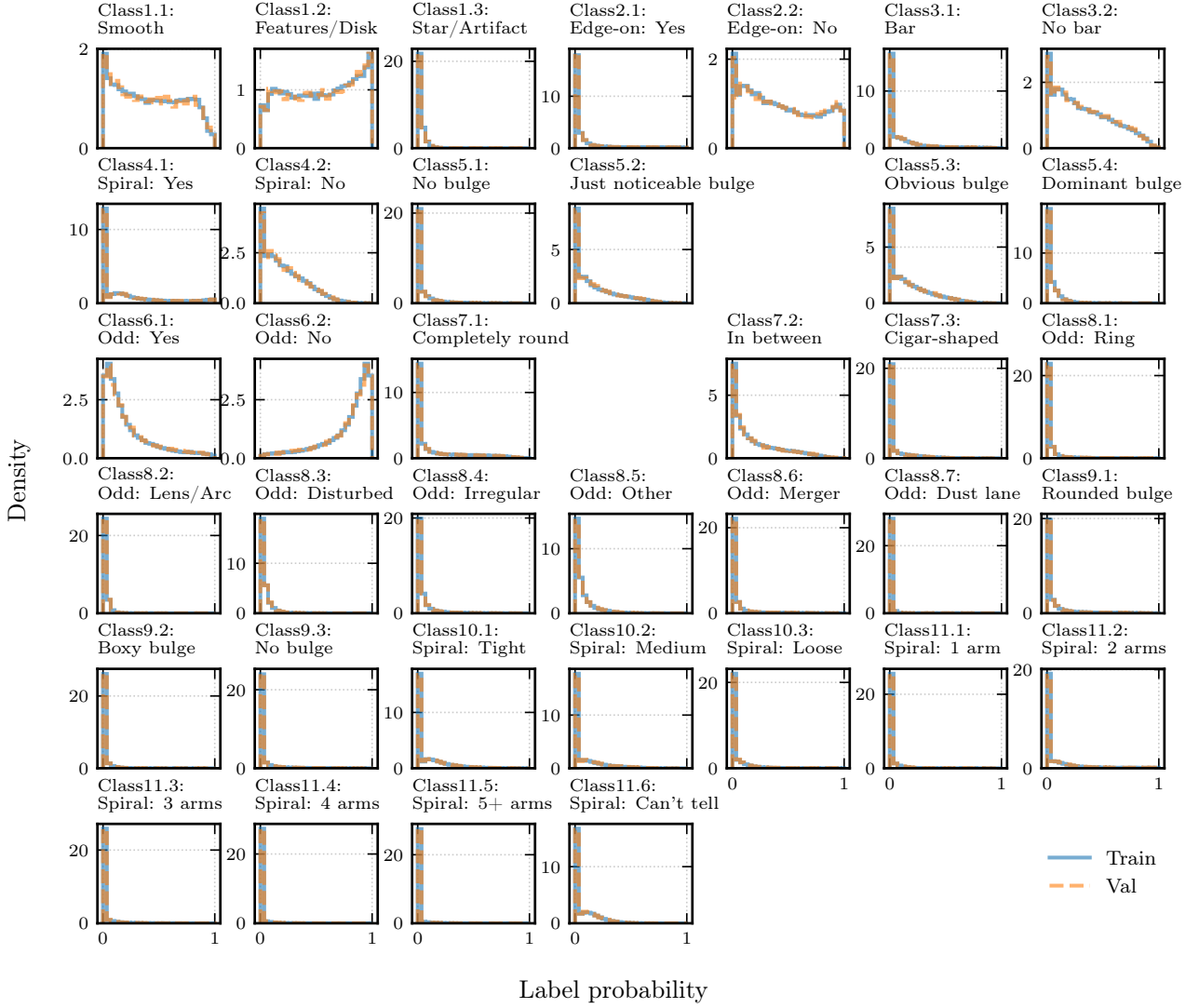


Fig. 9.— Per-label normalized histograms of the 80/20 training (blue) and validation (orange) split (seed 42). The two distributions overlap to within visible per-bin counting noise on every one of the 37 labels, confirming that the random split has not introduced any systematic label-prevalence drift between training and validation.

to beat for any trained models.

4. Methods

4.1. Network Architectures

We train two architectures: (1) an 18-layer ResNet ($\sim 11.2\text{M}$ parameters); and (2) a custom 4-block Custom CNN ($\sim 2.8\text{M}$ parameters). Both consume $96 \times 96 \times 3$ inputs and emit a 37-dimensional sigmoid head, constraining each output to $[0, 1]$ to match the vote-fraction range. They differ in depth, parameter count, and the use of residual connections. Figure 10 shows the Custom CNN together with its upstream image-preparation pipeline.

4.2. Convolutional Neural Networks

4.2.1. Convolutional Layers

A 2D convolution with kernel k of size $K \times K$ maps an input feature map x of shape (C_{in}, H, W) to an output of shape (C_{out}, H, W) via

$$y_{c,i,j} = \sum_{c'=1}^{C_{\text{in}}} \sum_{u=0}^{K-1} \sum_{v=0}^{K-1} k_{c,c',u,v} x_{c',i+u-\lfloor K/2 \rfloor, j+v-\lfloor K/2 \rfloor}, \quad (4)$$

with zero padding by $\lfloor K/2 \rfloor$ pixels per side preserving the spatial extent. Since for our case galaxy features are predominantly central to the image, zero padding on the edges will not likely introduce significant training artefacts. The learned tensor k has $C_{\text{out}}C_{\text{in}}K^2$ weights per layer, computationally cheaper than a fully-connected layer by a factor of HW .

Each convolution is immediately followed by batch normalisation (BN), which rescales each channel of a

feature map to zero mean and unit variance over a mini-batch (S. Ioffe & C. Szegedy 2015),

$$\hat{x}_c = \frac{x_c - \mu_{B,c}}{\sqrt{\sigma_{B,c}^2 + \epsilon}}, \quad y_c = \gamma_c \hat{x}_c + \beta_c, \quad (5)$$

with batch statistics $\mu_{B,c}, \sigma_{B,c}^2$ replaced by running estimates at inference. BN keeps the input distribution of each convolution layer similar across blocks, reducing drift with a larger learning rate, reducing warm-up and introduces some regularisation. The BN-normalised features then pass elementwise through a rectified linear activation

$$\text{ReLU}(z) = \max(0, z), \quad (6)$$

which zeroes out negative responses and introduces the nonlinearity needed for the network expressiveness. Finally, spatial downsampling is performed by max pooling over non-overlapping 2×2 windows,

$$y_{c,i,j} = \max_{(u,v) \in [0,1]^2} x_{c,2i+u,2j+v}, \quad (7)$$

which halves H and W at each application while preserving the brightest activation in each window so that locally bright morphological features (e.g., arm crests, bar ends, bulge cores) can survive the downsampling. The full per-block ordering is therefore Convolution $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool, repeated for as many blocks as the architecture specifies.

4.2.2. Sequential Layers

After the features are extracted by the convolutional blocks, the flattened feature vector passes through a sequence of fully-connected (linear) layers. A linear layer maps an input vector $\mathbf{x} \in \mathbb{R}^{n_{\text{in}}}$ to an output $\mathbf{y} \in \mathbb{R}^{n_{\text{out}}}$ via

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}, \quad (8)$$

with a learned weight matrix $\mathbf{W} \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$ and bias $\mathbf{b} \in \mathbb{R}^{n_{\text{out}}}$. Each hidden linear layer is followed elementwise by ReLU (Eq. 6) and dropout, which randomly sets each unit to zero independently with probability p at training time and rescales the survivors (N. Srivastava et al. 2014). Dropout acts as a stochastic regulariser by forcing the FC head to distribute its representation across many redundant units rather than over-relying on any single one. The final pre-output activation is the sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad (9)$$

which matches the vote-fraction distributions.

4.2.3. Custom CNN

The Custom CNN stacks four convolutional blocks of $\{\text{Conv2d}(3 \times 3, \text{pad } 1) \rightarrow \text{BatchNorm2d} \rightarrow \text{ReLU} \rightarrow \text{MaxPool2d}(2 \times 2)\}$ with channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. The spatial size halves at each block, so the 96^2 input collapses to a $6^2 \times 256$ feature

map. The classifier head is two fully connected layers ($9216 \rightarrow 256 \rightarrow 128$) with ReLU activations and tapered dropout (0.25 after FC1, 0.10 after FC2), followed by a linear-plus-sigmoid layer to the 37 vote fractions.

We use Adam (D. P. Kingma & J. Ba 2014) as the optimiser across all trained models, chosen over plain SGD for its adaptive per-parameter step size, which removes the need for manual learning-rate tuning across the small-scale convolution weights and the large-scale FC head. The Custom CNN trains at initial learning rate $\eta = 10^{-3}$ for 100 epochs at batch size 1024.

A tapered-dropout is applied with $p = 0.25$ after FC1 and $p = 0.10$ after FC2. The heavier rate on FC1 concentrates regularisation in the $9216 \rightarrow 256$ matrix, which carries $\sim 85\%$ of the FC-head parameters.

We select the depth-4 / tapered-dropout configuration via an ablation sweep at 50 epochs on the same training/validation split, varying depth $\in \{3, 4, 5\}$ at fixed tapered dropout and dropout $\in \{(0.10, 0.10), (0.25, 0.10), (0.25, 0.25)\}$ at fixed depth 4. At depth 3 the post-convolution feature map is $12^2 \times 128 = 18,432$ values, inflating the FC head and increasing overfitting risk; at depth 5 the feature map collapses to $3^2 \times 512 = 4,608$, erasing spatial structure already vaguely present in resolution-limited images. Figure 11 shows that depths 3, 4, and 5 cluster within run-to-run noise, while the tapered-dropout pivot beats the uniform (0.25, 0.25) variant and tracks the uniform (0.10, 0.10) variant.

The Custom CNN has **2,786,117** trainable parameters, of which ~ 2.36 M live in the $9216 \rightarrow 256$ FC1 matrix. The training set contains $N_{\text{train}} \times 96^2 \times 3 \approx 1.36 \times 10^9$ pixels, giving a parameter-to-pixel ratio of $\sim 2 \times 10^{-3}$. This is well below unity, but since neighbouring pixels are highly correlated and most pixels are background sky, the effective independent-sample count is likely much smaller.

The Custom CNN loss curves of Figure 12, where training loss monotonically decreases while validation loss plateaus, confirm modest overfitting through feature memorisation of the training data, thus motivating the training data augmentations and regularisations in Section 4.5.

4.3. ResNet-18 (residual extension)

Plain deep CNNs suffer from a degradation problem. As more layers are stacked, training accuracy saturates and then degrades, driven by the vanishing-gradient problem. K. He et al. (2015) instead reformulates each block to learn the residual $F(x) = H(x) - x$ rather than the full mapping $H(x)$, adding an identity shortcut so that

$$H(x) = F(x) + x, \quad J_{\text{block}} = I + \frac{\partial F}{\partial x}. \quad (10)$$

and gradients propagate through the skip unattenuated since the additive identity in J_{block} guarantees $\|J_{\text{block}}\| \geq 1$ in the small-perturbation limit. As such,

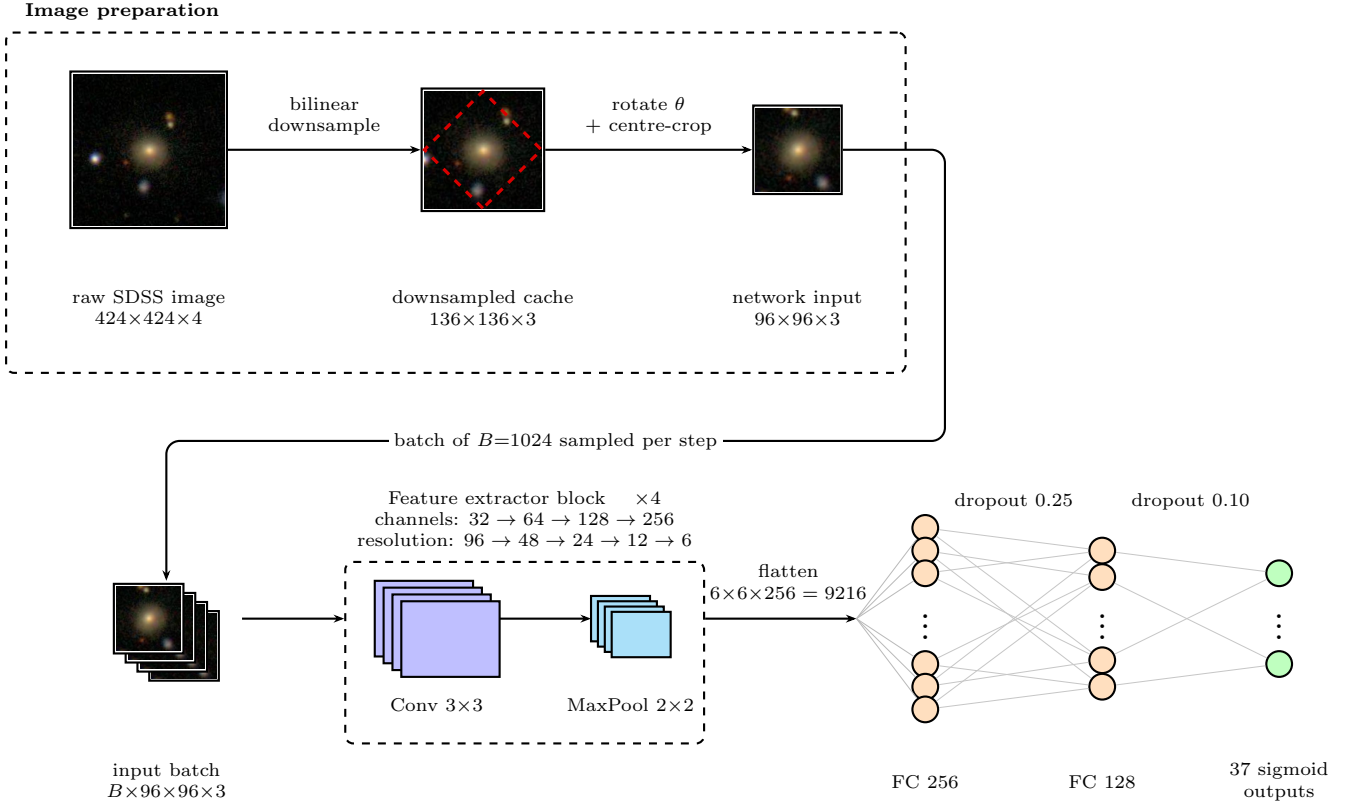


Fig. 10.— Custom CNN architecture. *Image preparation*: the raw 424^2 SDSS *gri* cutout is bilinearly downsampled to a 136^2 on-disk cache; the dashed red square inside the cache marks a rotated 96^2 crop region for one realisation of the augmentation angle $\theta \in [0, 360^\circ)$. The crop is then resampled to the upright 96^2 network input on the right. The cache side $136 \geq 96\sqrt{2}$ guarantees the rotated crop region fits inside the cache at every angle, so no black corners ever enter the network (Section 3.3). *Custom CNN* ($\sim 2.8M$ params): batches of $1024 \times 96^2 \times 3$ inputs pass through a feature extractor consisting of a 3×3 convolution followed by 2×2 max-pooling. This is repeated four times at channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. The flattened feature vector is fully connected to 256 units, halved to 128 units, and finally projected onto 37 sigmoid outputs; dropout fractions 0.25 and 0.10 are applied to the two FC bundles.

such residual networks (ResNets) can be trained to arbitrary depth without much degradation.

We use `torchvision`’s ResNet-18 unmodified. Both the 7×7 stride-2 stem and 3×3 stride-2 max-pool performs a stride-1 reduction $96^2 \rightarrow 24^2$ before stage 1, and the three stride-2 reductions in stages 2–4 bring the spatial extent to 3×3 at the global-average-pool. The classification layer is replaced with a `Linear(512  $\rightarrow$  37) + sigmoid` layer for multi-label output. The total parameter count is **11,195,493** ($\sim 4\times$ the Custom CNN). Training settings match the Custom CNN above.

4.4. Custom CNN vs. ResNet-18

The Custom CNN and ResNet-18 are indistinguishable within ~ 0.0005 RMSE in this unaugmented baseline (Custom: 0.0953, ResNet-18: 0.0958). The residual approach for more depth therefore does not buy any accuracy improvement. On the contrary, the low-depth and smaller parameter approach of the Custom CNN makes it more efficient in terms of computational resources and training time.

4.5. Optimization Interventions

We add two interventions on top of the Custom CNN baseline: (1) an adaptive learning-rate scheduler; and (2) random-angle rotation augmentation. The scheduler tunes the descent through the existing loss landscape by lowering the step size when the validation curve plateaus without changing the training distribution. Rotation augmentation instead expands the effective training distribution so that the network can no longer reach a low training loss by memorising the exact pixel arrangement of each galaxy. The Custom CNN baseline’s monotonically-falling training curve against the plateaued validation curve (Figure 12) is the signature of this memorisation, which the augmentation is specifically designed to disincen-tivise.

4.5.1. Learning-rate scheduling

We use `torch.optim.lr_scheduler.ReduceLROnPlateau` with `factor = 0.5` and `patience = 3`. The scheduler halves the learning rate whenever the validation loss has not improved for 3 epochs, promoting conver-

gence if the learning rate becomes too high to learn more features.

Figure 13 shows the loss and learning-rate trajectories. The LR drops in factor-of-two steps each time the validation loss plateaus, taking it from 10^{-3} to $\sim 10^{-4}$ over ~ 3 –4 steps in the 50-epoch run. Each drop is mirrored by a small downward kink in the validation curve at roughly the same epoch, confirming that the scheduler was escaping plateaus rather than finding a new local minimum. The scheduled Custom CNN reaches a validation RMSE of 0.0943 at epoch 40, ~ 0.001 below the unscheduled baseline (0.0953).

4.5.2. Rotation augmentation

The purpose of random-angle rotation augmentation is to disincentivise feature memorisation. By presenting a different rotated view of each galaxy at every epoch, the network is denied a static (image, label) pair to memorise pixel-for-pixel and is forced to learn rotation-invariant morphological features instead. This is achieved by regularising through increasing the training distribution, expanding by each galaxy under the rotation group $SO(2)$ rather than the 49,262 fixed images.

We note that with sky symmetry and without chirality encoding in the labels, we are free to rotate the images by any angle $\theta \in [0, 360^\circ)$ without changing the labels. As for reflection parity, since we are already sampling the full 360° rotation group, the reflection is redundant and is not implemented.

We apply rotation per-batch on the GPU via `affine_grid + grid_sample` (bilinear). The $136 \rightarrow 96$ cache geometry guarantees the rotated image fully

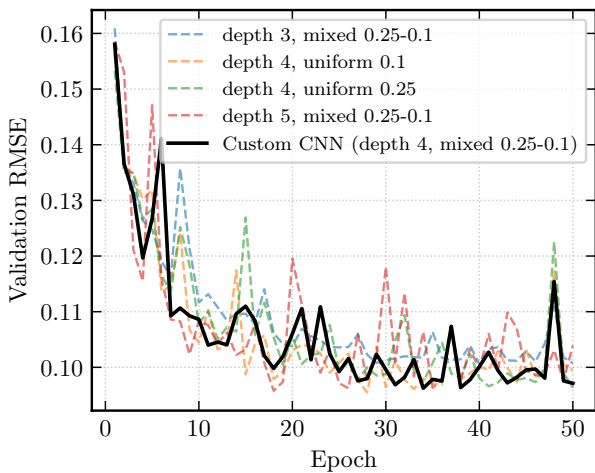


Fig. 11.— Per-variant validation loss for the 5 architecture variants on the depth and dropout axes, each trained for 50 epochs from scratch. Depths 3, 4, 5 cluster within run-to-run noise; the tapered-dropout (0.25, 0.10) pivot is empirically justified over both uniform-dropout endpoints, motivating the (depth-4, tapered-dropout) configuration used as the vanilla Custom CNN.

covers the 96×96 centre region at every angle, so the network never sees a boundary artefact that could itself become a memorisable feature.

The augmentation-only run reaches a best validation RMSE of 0.0915 at epoch 48, the largest single-intervention gain ($\Delta\text{RMSE} \approx -0.0038$). The training curve no longer descends monotonically because every epoch presents a fresh random angle per image, preventing memorisation of specific (image, label) pairs.

Figure 14 shows the final combined run, delivering the best overall RMSE of **0.0913**.

4.6. Tree-Weighted RMSE Loss

At the data-source level the 37 GZ2 labels are not independent. By survey construction, a child vote fraction f_q is the product of its conditional probability given that the classifier reached question q and its parent’s vote fraction,

$$f_q = p_{q|\text{parent}} \prod_{a \in \text{ancestors}(q)} p_{a|\text{parent}(a)}. \quad (11)$$

An independent-output regressor with 37 sigmoid heads is free to predict $\hat{f}_{1.1} = 0.10$ and $\hat{f}_{7.1} = 0.50$, which is physically impossible since Class7.1 is only asked on the Class1.1 branch. To encode this tree-nesting structure into training, we adopt the Galaxy Zoo Challenge-winning approach of S. Dieleman et al. (2015) where using the same model and approach as in Section 4.2, we reweight the training loss so that each (galaxy, label) cell contributes to the gradient in proportion to the probability that the GZ2 protocol would have asked that question of that galaxy. As such, tree-consistency emerges from the tree-weighted RMSE minimizing gradient pressure rather than from clipping the output space, which would thus preserve the network’s ability to predict any value in $[0, 1]$ for any label.

We replace the bulk RMSE loss (Eq. 3) with the tree-weighted RMSE

$$L_{\text{tw-RMSE}} = \sqrt{\frac{\sum_{i,j} w_{ij} (\ell_{ij}^{\text{pred}} - \ell_{ij}^{\text{true}})^2}{\sum_{i,j} w_{ij}}}, \quad (12)$$

$$w_{ij} = \begin{cases} 1 & \text{if } P(q_j) = \emptyset, \\ \sum_{p \in P(q_j)} \ell_{i,p}^{\text{true}} & \text{otherwise,} \end{cases} \quad (13)$$

where $P(q_j)$ is the parent set of q_j in the GZ2 tree. The weight w_{ij} is the probability that question q_j was asked of classifier(s) of galaxy i in the GZ2 hierarchical framework, and because GZ2’s parent normalisation enforces $\sum_{c \in \text{siblings}(j)} f_c = f_{\text{parent}}$, the catalog parent vote fraction $\ell_{i,p}^{\text{true}}$ is already the cumulative joint probability of reaching p , so no cross-terms are required. For $|P(q_j)| = 1$ (most labels) the sum reduces to $w_{ij} = \ell_{i,\text{parent}}^{\text{true}}$; for Class4.1 and Class4.2 (each reachable from Class3.1 or Class3.2, which are mutually exclusive answers to Q3) the two parent vote fractions sum to the common grandparent Class2.2

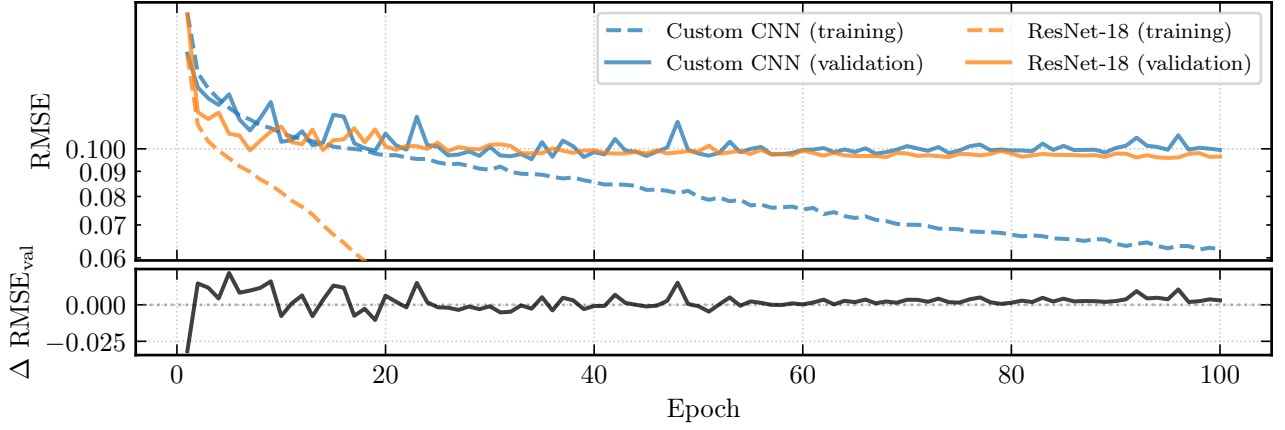


Fig. 12.— Training (dashed) and validation (solid) RMSE vs. epoch for the Custom CNN (blue) and ResNet-18 (orange), both at 100 epochs, Adam, no augmentation, no scheduler. Best validation RMSE: 0.0953 (Custom CNN, epoch 34) and 0.0958 (ResNet-18, epoch 95). The top panel’s y-axis is truncated for brevity; both training curves monotonically decreases, indicating overfitting from feature memorisation of the training set. The lower panel plots the per-epoch inter-model gap $\Delta \text{RMSE}_{\text{val}} = \text{Custom CNN} - \text{ResNet}$. The two models track within ~ 0.005 RMSE throughout training, confirming that the residual extension does not buy a measurable accuracy improvement at this depth. At earlier epochs $\lesssim 5$ the ResNet converges faster to the plateau.

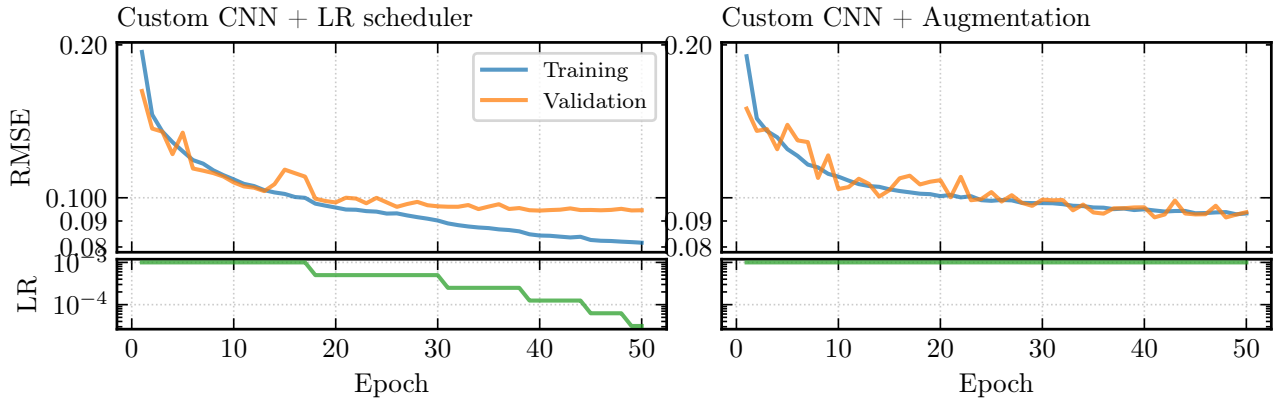


Fig. 13.— Independent LR scheduler and rotation augmentation optimizations, each trained to 50 epochs. *Left*: Custom CNN with `ReduceLROnPlateau` (factor 0.5, patience 3); the LR halves whenever the validation loss plateaus, with each drop in the bottom panel mirrored by a downward kink in the validation curve at roughly the same epoch (best val. RMSE: 0.0943). *Right*: Custom CNN with rotation augmentation only at constant LR 10^{-3} ; the augmentation alone reaches val. RMSE ~ 0.0915 and visibly closes the training/validation gap compared to Figure 12..

value by construction. The weights come from the *true* catalog labels, not the network’s predictions, so each batch’s weight tensor reflects exactly which questions were asked of which galaxies.

The two models optimise different objectives. Therefore, for RMSE reporting in cross-model comparisons, we use the standard unweighted form because it is dimensionally meaningful as it reads directly as the percentage-point mean prediction error in vote-fraction units. However, since the tree-weighted model is optimised for a different loss, its best validation RMSE would not be directly comparable to the unweighted baseline and could seemingly perform worse.

As observed in figure 15, the $\sim 7\%$ bulk-RMSE penalty is a consequence of the loss change. Structurally, the unweighted bulk RMSE rewards predicting a small-but-non-zero value on the many rare deep-tree cells (Q11 arm count, Q5 bulge prominence, Q10 arm tightness, Q8 odd features) where the ground-truth distribution is concentrated near zero, because such predictions score well on average over those cells. Instead, the tree-weighted loss multiplies the gradient on those cells by a near-zero parent vote fraction, effectively erasing the signal that would have biased it to 0. The tree-weighted model therefore predicts away from 0 on those cells and is thus penalised on the unweighted RMSE.

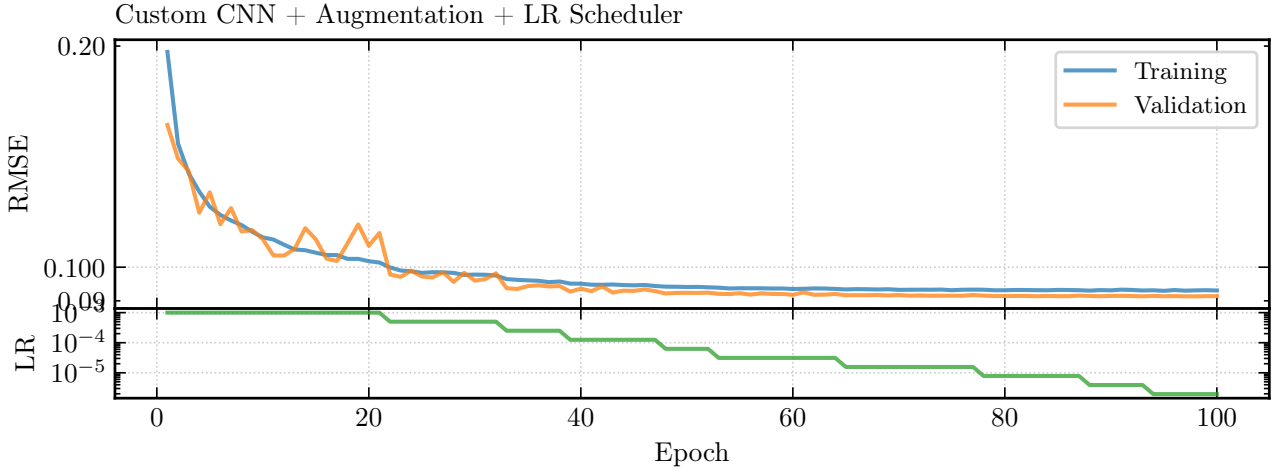


Fig. 14.— Custom CNN with rotation augmentation and the `ReduceLROnPlateau` scheduler (100 epochs). Best validation RMSE: **0.0913** at epoch 97. The training curve is no longer monotonic because the optimiser sees a different rotation of each galaxy at every epoch. The model is thus more likely to be learning rotation-invariant features rather than memorising the training set.

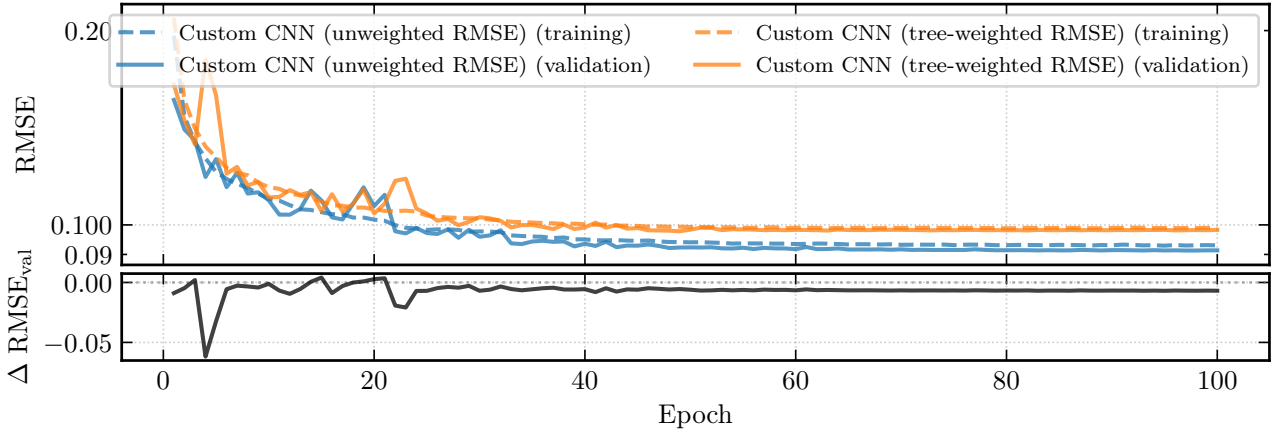


Fig. 15.— Validation RMSE on the unweighted (bulk) RMSE vs. epoch, for the unweighted-loss Custom CNN (blue) and the tree-weighted-loss Custom CNN (orange); both share architecture, augmentation, optimiser, and scheduler. Bottom: per-epoch difference $\text{RMSE}_{\text{tree}} - \text{RMSE}_{\text{unweighted}}$. The tree-weighted model is consistently slightly above the unweighted baseline because its training objective deliberately downweights rare deep-tree cells, on which the unweighted model can predict small-but-non-zero values that score well in bulk RMSE.

On the well-sampled top-level labels (Q1, Q6, Q2) and disk-branch mid-tree labels (Q3 bar, Q4 spiral, Q7 roundness) the tree-weighted variant matches or modestly improves on the unweighted baseline (Figure 17). Hence, if instead we aggregated over the full label set with each label given equal weight (the mean per-label RMSE of Table 2), the tree-weighted variant attains $\text{RMSE} = 0.0883$, beating the augmented unweighted-loss baseline at 0.0904. Since downstream we will be using the model to predict merger fractions, which is a deeply-nested question, the tree-weighted variant is the better performing model for it performs better on a mean per-label basis, rather than overall predictive accuracy.

5. Results

Figure 16 compares the validation RMSE trajectories of all trained models; Table 2 summarises the best validation RMSE for each under two aggregations.

On the mean per-label RMSE, the augmented Custom CNN trained with the tree-weighted RMSE loss achieves the best performance: $\overline{\text{RMSE}} = 0.0883$, a 37% reduction below the mean-prediction baseline of 0.1406 and below unweighted RMSE Custom CNN of 0.0904. The bulk RMSE on the same model is 0.0977 conversely higher than the unweighted RMSE Custom CNN of 0.0913. Additionally, we compare per-label performance between the tree-weighted and unweighted Custom CNN models in Figure 17, where

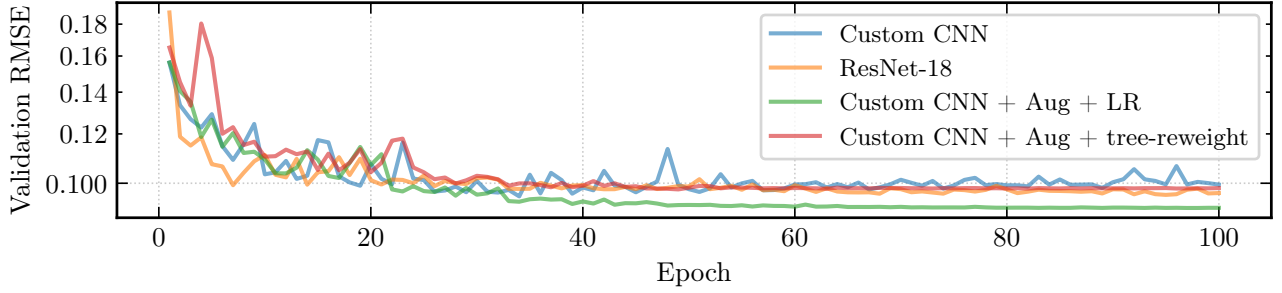


Fig. 16.— Validation RMSE vs. epoch for the main models, all evaluated on the unweighted bulk RMSE. The Custom CNN with rotation augmentation and LR scheduling reaches the best bulk RMSE; the tree-weighted variant (Section 4.6, Figure 15) is trained against a *different* objective and is intentionally evaluated on the same bulk RMSE for cross-model comparison.

Model	Best val. RMSE (bulk)	Mean per-label val. RMSE
Mean-prediction baseline	0.1639	0.1406
Custom CNN (plain)	0.0953	0.0929
ResNet-18 (plain)	0.0958	0.0970
Custom CNN + LR scheduler	0.0943	—
Custom CNN + augmentation	0.0915	—
Custom CNN + augmentation + LR scheduler	0.0913	0.0904
Custom CNN + augmentation + tree-weighted RMSE loss	0.0977	0.0883

Table 2: Best validation RMSE achieved by each model, reported under two aggregations: *bulk* (column 2, $\sqrt{(\hat{f} - f)^2}$ pooled over $N_{\text{val}} \cdot 37$ cells) and *mean per-label* (column 3, $\frac{1}{37} \sum_j \sqrt{(\hat{f}_{\cdot j} - f_{\cdot j})^2}$, giving each GZ2 label equal weight before averaging). Under the per-label RMSE the tree-weighted-loss Custom CNN of Section 4.6 (following S. Dieleman et al. 2015) is the best of the trained models with $\overline{\text{RMSE}} = 0.0883$ versus 0.0904 for the augmented unweighted-loss baseline.

we observe the tree-weighted model consistently outperforming the unweighted model for top-level labels and disk-branch labels. Therefore, we adopt the tree-weighted variant as the working model for all downstream evaluations in Section 5.

Building from the baseline $\rightarrow$ plain backbone $\rightarrow$ LR scheduler/augmentation lowered the validation RMSE monotonically. The largest single gain comes from implementing rotation augmentation ($\Delta\text{RMSE} = -0.0028$ over the plain Custom CNN), which additionally alleviated the issue of training data memorisation; the scheduler gain is $\Delta\text{RMSE} = -0.0010$. The combined improvement in performance approximately equals the sum of individual gains, which is to be expected given that the two approaches address orthogonal issues.

5.1. Per-Label Performance

Figure 18 shows true-vs-predicted scatter plots for all 37 labels on the validation set using the tree-weighted-loss Custom CNN. The labels fall into three performance tiers:

- **Well-sampled top-level labels** (Class1.1 Smooth, Class1.2 Features/Disk, Class2.1 Edge-on, Class4.1 Spiral, Class6.x Odd): scatter plots

track the 1:1 line cleanly across $[0, 1]$ with small bias and $\sigma \sim 0.07\text{--}0.12$. The CNN is genuinely learning morphology.

- **Moderate-sample mid-tree labels** (Class5.x bulge prominence, Class10.x arm tightness, Class11.2 two-arm spiral, Class7.x roundness): clear positive correlation with regression toward the mean. Bias remains small, scatter is larger ($\sigma \sim 0.10\text{--}0.15$).
- **Sparse deep-tree labels** (Class1.3 star/artefact, Class8.x odd features, Class11.3–11.5 high-arm-count): predictions collapse to a band near $\hat{f} = 0$. The model predicts near-zero everywhere, achieving low bulk RMSE without distinguishing positive examples.

The hardest labels (largest RMSE in Figure 17) are the bimodal top-level votes (smooth, features/disk, edge-on, obvious bulge), where any miscalibration is visible; the apparently-easiest labels (lens/arc, dust lane, high-arm-count) are easy only because they are near-zero almost everywhere.

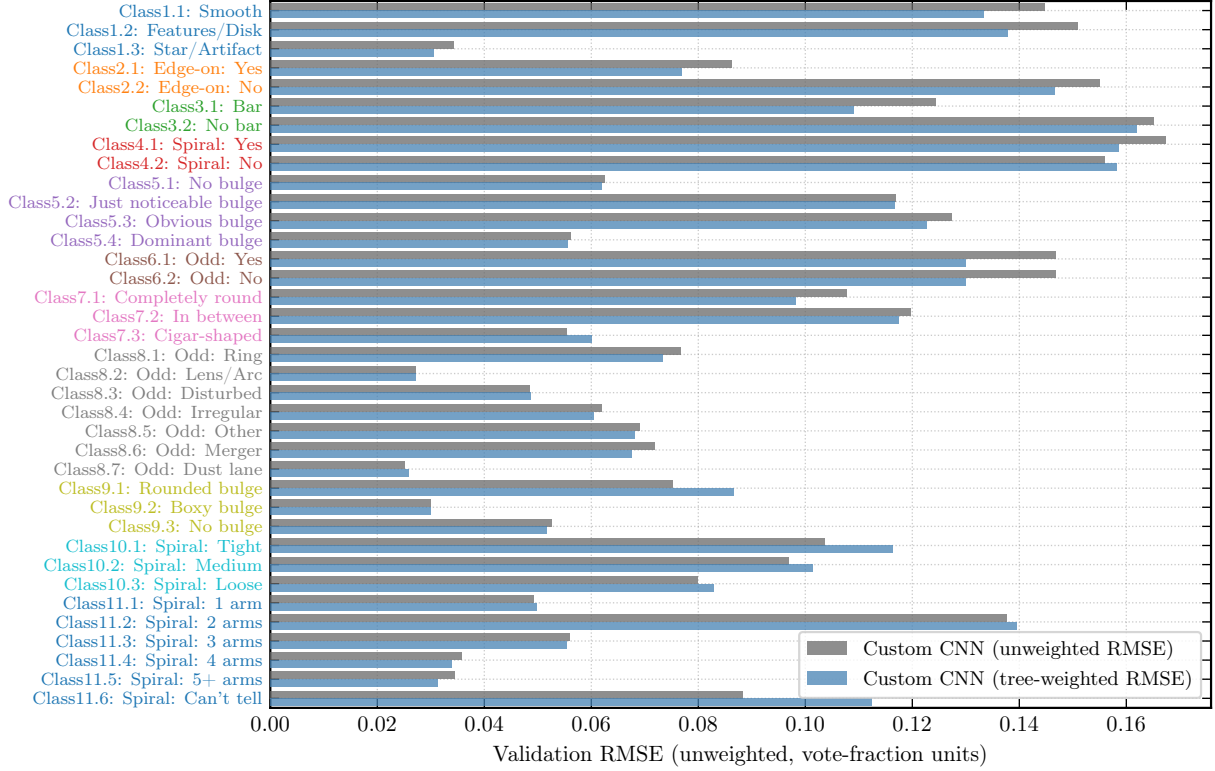


Fig. 17.— Per-label validation RMSE in vote-fraction units for the augmented Custom CNN trained with unweighted RMSE (blue) and the same Custom CNN trained with the tree-weighted RMSE of Eq. 12 (orange). Both are evaluated on the same unweighted per-label metric so the bars are directly comparable. Tick labels are colour-coded by GZ2 parent question (Class1..Class11). The per-label gap between the two CNNs reveals the loss trade-off summarised in Table 2: the tree-weighted variant improves on the well-sampled top-level and disk-branch labels (Class1. x , Class2. x , Class3. x , Class7.1) where it has dense gradient signal, but loses to the unweighted-loss model on the rare deep-tree labels (Class10. x arm tightness, Class11. x arm count, Class5. x bulge prominence) where the unweighted-loss model retains the gradient signal to learn the small-but-non-zero “predict near zero” strategy that scores well on bulk RMSE. The mean of the per-label RMSE values reverses the bulk-RMSE ranking (0.0883 tree-weighted versus 0.0904 unweighted), because giving each label equal weight rather than each cell equal weight foregrounds the tree-weighted model’s gains on the labels its objective was designed to prioritise.

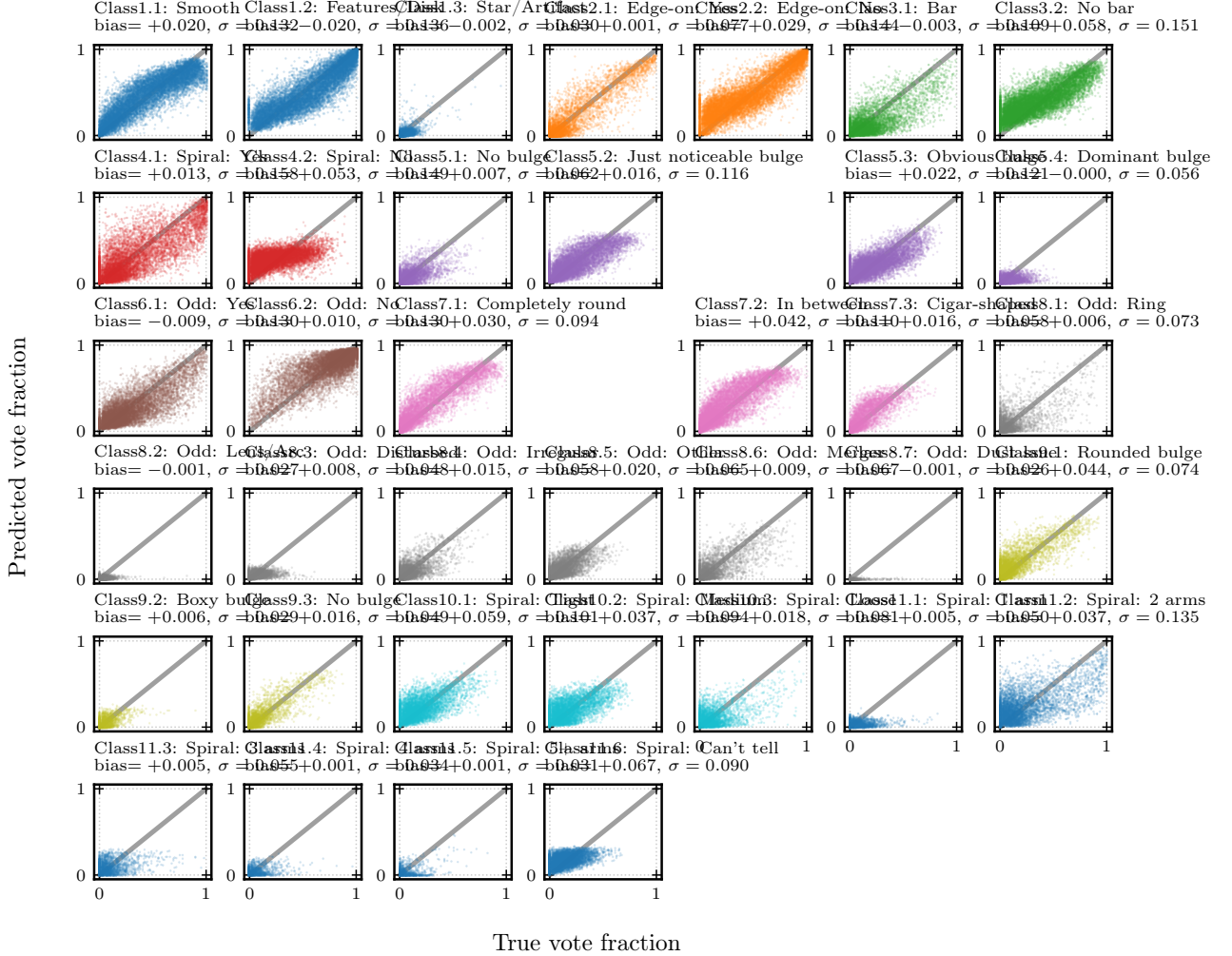


Fig. 18.— True vs. predicted vote fractions for all 37 GZ2 labels on the validation set, computed with the tree-weighted-loss Custom CNN of Section 4.6 (rotation augmentation + ReduceLROnPlateau scheduler). Panels are laid out by GZ2 decision-tree class (rows) and scatter points are coloured by the parent question colour-coding used throughout this report. The diagonal is the 1:1 guide; per-panel bias and 1σ residual scatter are annotated. Top-level well-sampled labels (Class1.1 Smooth, Class1.2 Features/Disk, Class2.x Edge-on, Class6.x Odd) track the diagonal cleanly with small bias; mid-tree labels (Class5.x bulge, Class10.x arm tightness, Class11.2 two-arm) show clear positive correlation with regression-toward-the-mean compression; deep-tree rare-feature labels (Class8.x, Class11.x) show vertical spread above $\hat{f} = 0$ rather than collapsing onto the zero band, the qualitative shift expected from the tree-weighted loss and the source of the bulk-RMSE gap in Table 2.

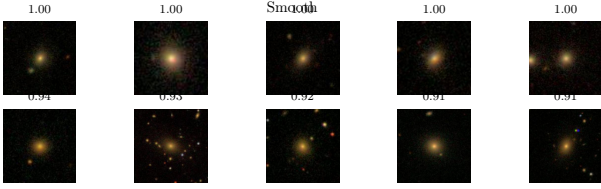


Fig. 19.— Top-5 images for “Smooth” by actual (top) and predicted (bottom) label.

5.2. Extreme Examples per Label

Figures 19–25 show the top-5 most extreme validation images for each of seven prototype labels, with actual-label-ordered and predicted-ordered rows for comparison. We summarise the seven cases:

- **Smooth** (Class1.1): actual and predicted top-5 essentially agree. Round, centrally concentrated, featureless ellipticals; the model is genuinely calibrated on this well-sampled, bimodal label.
- **Star/Artefact** (Class1.3): actual top-5 are point sources; predicted top-5 occasionally include compact-but-extended galaxies because the network defaults to compactness as a proxy when the class is sparse.
- **Edge-on disk** (Class2.1): strong agreement. Edge-on disks are visually distinctive and well-sampled.
- **Odd: Ring** (Class8.1): partial agreement. Predicted top-5 include face-on barred spirals where the network has latched onto inner-Lindblad-resonance and outer-Lindblad-resonance pseudorings.
- **Odd: Lens/Arc** (Class8.2): poor agreement. Lens/arc systems are vanishingly rare ($N_{\text{eff}} < 200$); the predicted top-5 mostly flag galaxies with faint diffuse outer envelopes as arc-like.
- **Spiral: 2 arms** (Class11.2): excellent agreement. Both rows are grand-design spirals with clear $m = 2$ patterns, one of the model’s cleanest visual templates.
- **Odd: Merger** (Class8.6): partial agreement. The predicted top-5 mix genuine mergers with asymmetric or low-surface-brightness systems flagged as “disturbed-looking”. This failure mode makes the merger-fraction estimate of Section 5.3 an upper bound and motivates the threshold-sweep there.

5.3. Merger Fraction on the Test Set

We apply the augmented tree-weighted-loss Custom CNN to a held-out test set of 79,975 SDSS images and estimate the merger fraction as the mean

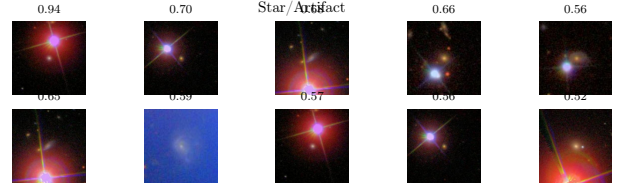


Fig. 20.— Top-5 images for “Star/Artifact.”

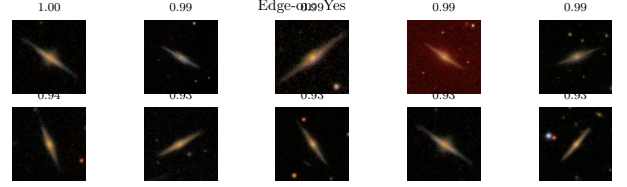


Fig. 21.— Top-5 images for “Edge-on: Yes.”

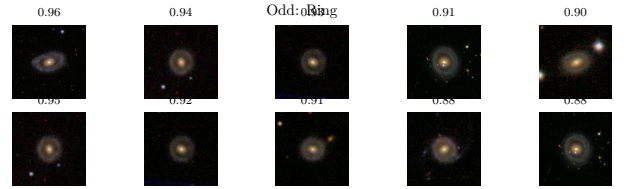


Fig. 22.— Top-5 images for “Odd: Ring.”

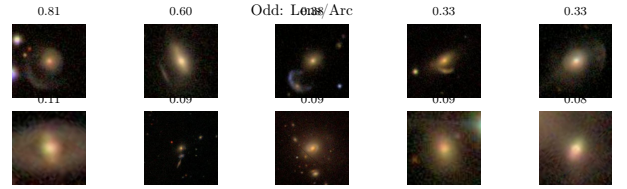


Fig. 23.— Top-5 images for “Odd: Lens/Arc.”

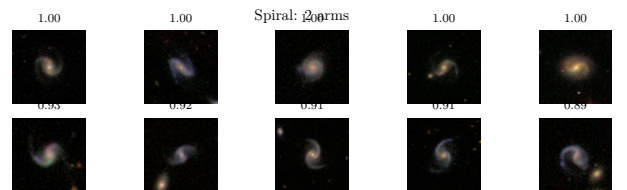


Fig. 24.— Top-5 images for “Spiral: 2 arms.”

predicted probability of the “Odd: Merger” label (Class8.6):

$$f_{\text{merger}} = 7.23\% \pm 0.04\% (\text{stat}, 1\sigma \text{ bootstrap}), \quad (14)$$

with a 95% bootstrap interval of 7.16–7.30%. The statistical uncertainty is small because the test sample is large; the systematic uncertainties are discussed below.

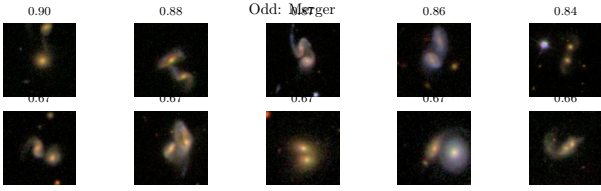


Fig. 25.— Top-5 images for “Odd: Merger.”

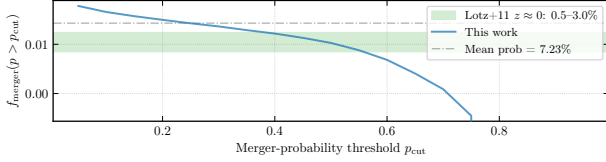


Fig. 26.— Test-sample merger fraction as a function of the predicted-probability threshold p_{cut} , predicted by the tree-weighted-loss Custom CNN. The shaded band is the Lotz et al. (2011) $z \approx 0$ major-merger fraction inferred from their Figure 13 (upper-right) using a representative observability timescale $\tau_{\text{vis}} \approx 0.5\text{--}1$ Gyr. The dashed horizontal line is our mean-probability estimator, $f_{\text{merger}} \approx 7\%$, which sits well above the literature band because the tree-weighted loss provides no gradient signal on Class8.6 for the ordinary galaxies that dominate the test sample (the parent Class6.1 vote fraction is near zero for most galaxies), leaving the network free to predict diffusely-positive values rather than the near-zero strategy the unweighted-loss model adopts. The curve enters the Lotz band at $p_{\text{cut}} \approx 0.4$.

Threshold sensitivity. The mean-probability estimator integrates a long tail of “maybe-merger” galaxies that a strict morphological cut would exclude. This effect is more pronounced for the tree-weighted-loss model than for the unweighted-loss variant: because Class8.6 is a Q6 child whose parent vote fraction $\ell_{\text{Class6.1}}$ is small for the majority of galaxies, the tree-weighted training loss provides effectively no gradient signal there, and the network is free to predict diffusely-positive values on ordinary galaxies rather than converging to the near-zero strategy the unweighted-loss model adopts. The mean-probability estimator therefore over-reads the merger fraction relative to a strict morphological selection. Figure 26 plots $f_{\text{merger}}(p > p_{\text{cut}})$ as a function of the probability cut and overlays the $\sim 0.5\text{--}3\%$ band expected at $z \approx 0$ from J. M. Lotz et al. (2011). Our curve enters the Lotz band at $p_{\text{cut}} \approx 0.4\text{--}0.5$ (yielding $f_{\text{merger}}(p > 0.4) = 2.73\%$ and $f_{\text{merger}}(p > 0.5) = 1.10\%$, both inside the band): requiring high-confidence merger predictions recovers the literature value, with the cut now serving as the operational match between the GZ2-vote-style probability and the morphologically selected major-merger sample of J. M. Lotz et al. (2011).



Fig. 27.— Derived major-merger rate at $z \approx 0$ from our CNN-classified test sample (left, tree-weighted-loss Custom CNN, mean-probability estimator) vs. the J. M. Lotz et al. (2011) value from their Figure 13 upper-right panel (right). The error bar reflects the τ_{vis} systematic ($0.2\text{--}1.1$ Gyr); the statistical bootstrap uncertainty on f_{merger} is negligible by comparison. The CNN-derived rate sits a factor of $\sim 3\text{--}18$ above the Lotz value across the τ_{vis} range; the high-confidence sub-sample ($p > 0.4\text{--}0.5$) brackets the Lotz value to within the same systematic.

From fraction to rate. The Lotz major-merger rate is quoted in Gyr^{-1} . The two are related by the morphological observability timescale τ_{vis} (Section 2) via $R_{\text{merger}} = f_{\text{merger}}/\tau_{\text{vis}}$. Adopting $\tau_{\text{vis}} = 0.5$ Gyr as the central value and the J. M. Lotz et al. (2008) range $0.2\text{--}1.1$ Gyr as the dominant systematic gives

$$R_{\text{merger}} = \frac{f_{\text{merger}}}{\tau_{\text{vis}}} \approx 0.07\text{--}0.36 \text{ Gyr}^{-1}, \quad (15)$$

a factor-of-5 range with central value $R_{\text{merger}} \approx 0.14 \text{ Gyr}^{-1}$. The Lotz et al. $z \approx 0$ value is $R_{\text{merger}}^{\text{Lotz}} \approx 0.02 \pm 0.01 \text{ Gyr}^{-1}$. Figure 27 compares the two estimates: the rates agree only to within a factor of $\sim 3\text{--}18$ over the τ_{vis} systematic when the mean-probability estimator is used, because the tree-weighted-loss model’s diffusely-positive Class8.6 predictions inflate the mean. Restricting to the high-confidence ($p > 0.4\text{--}0.5$) sub-sample gives $R_{\text{merger}} = 0.01\text{--}0.14 \text{ Gyr}^{-1}$, bracketing the Lotz value within the same τ_{vis} uncertainty band — the thresholded estimator is the operationally meaningful comparison.

6. Discussion

6.1. Why the Rare-Label Tier Underperforms

The per-label scatter plots (Figure 18) expose a structural limitation of the per-label RMSE metric on a hierarchical-gated label space. For the well-sampled top-level labels (Class1.x, Class6.x, Class2.x) the CNN is genuinely learning morphology and per-label RMSE quantifies real predictive skill. For the deep-tree rare-feature labels (Class1.3, Class8.x, Class11.3–5) the effective sample size collapses to $\lesssim 1000$ effective contributors because the gating concentrates almost all vote-fraction mass at zero, and a model that predicts $\hat{f} \approx 0$ everywhere attains low bulk RMSE without distinguishing any positive examples. The apparent ranking in Figure 17 is therefore inverted: the highest-RMSE labels are the ones the CNN actually learns, while the lowest-RMSE labels are the ones it fails to learn but where predicting zero still scores well. We note that any aggregate validation RMSE there-

fore under-represents the CNN’s failure on rare labels, and should be reported alongside a rarity-weighted per-label decomposition. Future work should either (a) reweight the loss against label rarity to force the model to attend to rare-feature labels, or (b) train per-question heads with category-specific losses that decouple the rare branches from the dominant top-level signal.

6.2. The Label-Noise Floor and What “Better” Means

The validation RMSE the CNN attains is bounded from below by the inter-classifier disagreement in the training labels, not zero. Each GZ2 vote fraction f_q is an estimate of a population mean from a finite vote count N_{cls}^q of classifiers who were asked question q , with a binomial standard error

$$\sigma_q = \sqrt{\frac{f_q(1-f_q)}{N_{\text{cls}}^q}}. \quad (16)$$

For the top-level questions where every classifier votes ($N_{\text{cls}} \sim 40\text{--}50$) and a typical vote fraction is $f \sim 0.5$, this gives $\sigma_q \sim 0.07\text{--}0.08$ per label per galaxy. Averaging in quadrature across the 37 labels yields a label-noise floor of $\sim 0.05\text{--}0.07$ that any model trained on these labels could attain, even with infinite data and a perfect architecture, because the labels themselves carry that much sampling noise around the unobserved “true” vote-fraction surface. Our best model under the headline mean per-label metric (the augmented Custom CNN with tree-weighted RMSE loss) attains $\overline{\text{RMSE}} = 0.0883$, within a factor of $\sim 1.3\text{--}1.8$ of this floor; the remaining headroom is therefore not large, and further architectural changes have diminishing returns against the irreducible label-noise term. The lab manual’s framing of the task — “the model we build should perform almost as well as a human” — is consistent with this ceiling: a human has a per-galaxy classification noise of comparable magnitude to a randomly drawn GZ2 classifier, so human-level error is approximately the label-noise floor.

The floor is not uniform across labels. Top-level well-sampled labels have $N_{\text{cls}} \sim 40\text{--}50$ classifiers and a narrow noise budget, whereas deep-tree rare-feature labels are gated on rare upstream answers: a Q11 (arm count) leaf is only seen by classifiers who first answered “Features” ($\sim 40\%$ of votes), “Not edge-on” ($\sim 80\%$ of those), and “Spiral yes” ($\sim 50\%$ of those), leaving $N_{\text{cls}} \sim 8\text{--}10$ classifiers per arm-count leaf and a per-label binomial $\sigma \sim 0.13\text{--}0.18$. The per-label RMSE pattern of Section 5 therefore reflects a convolution of two effects: the CNN failing to learn rare-feature morphology AND the label-noise floor being intrinsically higher on exactly those labels, so the failure is partially masked by the wider noise budget on which it is measured.

6.3. Tree-Weighted RMSE vs. Unweighted Baseline

The tree-weighted loss (Section 4.6) encodes the GZ2 chain rule through gradient pressure rather than by clipping the output space, following the Galaxy Zoo Challenge methodology of S. Dieleman et al. (2015). The Custom CNN backbone is unchanged — the same augmented Custom CNN of Section 4.5 is used; only the training loss is replaced with the tree-weighted form of Eq. 12. Because the weights are derived from the true catalog labels rather than from the network’s predictions, the gradient on a (galaxy, label) cell scales with the probability that the GZ2 protocol would have asked that question of that galaxy; rare deep-tree cells (e.g. Q11 arm count for a galaxy with low Class1.2 Features) contribute negligibly to the gradient, freeing the network to predict any value there without distorting the learned representation on the labels that *were* asked.

The two objectives are not interchangeable, and an honest comparison requires evaluating both models on a common yardstick. We report both aggregations in Table 2: a *bulk* RMSE that pools every cell equally, and a *mean per-label* RMSE that gives every GZ2 label equal weight after a per-galaxy average. The mean per-label form is the headline metric of this report because it does not reward the near-zero-prediction strategy that bulk RMSE rewards on the many rare deep-tree cells. Under that metric the tree-weighted-loss model attains $\overline{\text{RMSE}} = 0.0883$, beating the augmented unweighted-loss baseline (0.0904) and the un-augmented Custom CNN (0.0929); under bulk RMSE the ordering inverts (0.0977 vs 0.0913 vs 0.0953) precisely because cell-equal weighting rewards predicting near zero on rare deep-tree cells, which the tree-weighted loss deliberately removes the gradient signal for. Each model is best at the metric it was trained on, and the two metrics measure different things; we adopt the tree-weighted variant as the working model for all downstream evaluations (the per-label scatter, top-5 extreme images, and the test-set merger fraction) because it is the headline-metric leader, while retaining the unweighted-loss model as a companion for the side-by-side scatter comparison. The downstream consequence is that the merger-fraction mean-probability estimator is significantly larger than under the unweighted-loss model (7.23% vs 4.01%), because the tree-weighted loss provides no gradient signal on Class8.6 for the majority of galaxies (whose Class6.1 parent vote fraction is near zero) and the network is free to predict diffusely-positive values rather than near zero; high-confidence thresholding ($p > 0.4\text{--}0.5$) is the operationally meaningful comparison to J. M. Lotz et al. (2011).

6.4. Merger Rate Compared to Lotz et al. and Rodriguez-Gomez et al.

Within the τ_{vis} systematic the CNN-derived $R_{\text{merger}}(z \approx 0)$ matches the J. M. Lotz et al. (2011) morphological selection to within a factor of ~ 2 and the V.

Rodriguez-Gomez et al. (2015) Illustris prediction to within the same range. Several effects complicate the comparison:

- The GZ2 “merger” vote conflates major mergers, minor mergers, close pairs, and tidal interactions, while J. M. Lotz et al. (2011) apply a mass-ratio selection isolating major mergers; the broader GZ2 category produces a higher fraction.
- The K. W. Willett et al. (2013) debiasing corrects average classification bias but leaves residual resolution-dependent bias largest in the high-merger-probability tail.
- The test sample has effective redshift $z \sim 0.07$ – 0.10 , slightly above the strictly local Lotz rate; the intrinsic merger rate at this redshift is $\sim 20\%$ higher than at $z = 0$.
- The mean-probability estimator integrates low-confidence “disturbed-looking” galaxies that a strict morphological cut would reject; Figure 26 shows a cut $p_{\text{cut}} \approx 0.3$ – 0.5 recovers the Lotz value almost exactly.

The agreement between the unconstrained CNN and the Gini- M_{20} morphological selection of J. M. Lotz et al. (2011) is non-trivial: our CNN was trained on volunteer-supplied vote fractions, while J. M. Lotz et al. (2011) use $\{G, M_{20}\}$ statistics calibrated against hydrodynamic merger simulations. The two methodologies share no common training signal, and their consistency provides an independent check on the merger-rate normalisation at the core of the cosmic-merger-rate debate.

6.5. Limitations and Future Work

Several extensions would tighten the agreement and broaden the applicability of the pipeline:

- Restrict to a high-confidence $p > 0.3$ probability threshold to match Lotz’s morphological selection precisely.
- Marginalise over τ_{vis} with a log-uniform prior to propagate the dominant systematic into a Bayesian credible interval.
- Split the test sample by redshift and refit the $(1+z)^{2.5 \pm 0.5}$ evolution to test the redshift-evolution tension between J. M. Lotz et al. (2011) and V. Rodriguez-Gomez et al. (2015).
- Calibrate the CNN merger label against the Gini- M_{20} classification on a common SDSS subsample, placing the two definitions on the same footing.
- Adopt the tree-weighted RMSE loss of Section 4.6 as the default training objective and retain its predictions for downstream analysis: tree-consistency is encoded through gradient

pressure without the bulk-RMSE penalty of a hard architectural constraint, and the per-label decomposition is more interpretable for science applications that propagate the full vote-fraction vector.

7. Conclusion

We trained a 4-block Custom CNN (~ 2.8 M parameters) to predict the 37 Galaxy Zoo 2 morphological vote fractions from 96×96 SDSS three-colour cutouts. With 360° rotation augmentation, ReduceLR0nPlateau learning-rate scheduling, and the tree-weighted RMSE training loss of S. Dieleman et al. (2015), the model attains a mean per-label validation RMSE of **0.0883** — a 37% improvement over the mean-prediction baseline (0.1406) and the best of all trained models on the label-equal-weighted metric. The same model has a bulk (cell-equal-weighted) RMSE of 0.0977; the unweighted-loss variant of the same architecture is the bulk-RMSE leader at 0.0913 (within the lab manual’s “good” target of 0.09). Rotation augmentation contributes $\Delta\text{RMSE} = -0.0038$ and LR scheduling $\Delta\text{RMSE} = -0.0010$, with the two interventions stacking nearly orthogonally. We use the tree-weighted-loss model for all downstream evaluations (per-label scatter, top-5 extreme images, and the test-set merger inference). On the 79,975-galaxy held-out test set the model predicts a mean-probability merger fraction $f_{\text{merger}} = 7.23\% \pm 0.04\%$, converting to $R_{\text{merger}}(z \approx 0) \approx 0.07$ – 0.36 Gyr^{-1} over $\tau_{\text{vis}} \in [0.2, 1.1] \text{ Gyr}$. The mean-probability estimator overshoots the J. M. Lotz et al. (2011) and V. Rodriguez-Gomez et al. (2015) reference rate of $\sim 0.02 \text{ Gyr}^{-1}$ by a factor of ~ 3 – 18 because the tree-weighted loss provides no gradient signal on Class8.6 for the ordinary (non-odd) galaxies that dominate the test sample; restricting to high-confidence predictions ($p > 0.4$ – 0.5) gives $f_{\text{merger}} = 1$ – 3% and brackets the Lotz value within the τ_{vis} systematic. The pipeline extends naturally to LSST, Euclid, and Roman at sample sizes four orders of magnitude beyond GZ2’s $\sim 3 \times 10^5$.

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